THE Q-MODEL — From Q to Cosmology

A Research Monograph on Structure, Complexity and Random Actualization

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Abstract

This monograph is the research-monograph edition of THE Q-MODEL (TQM), a theory of structure and content that compresses observable physics to four primitives — the individuation principle Q, Random Actualization, the scale triad $(\ell, \tau, \hbar)$, and a single continuous nonlinearity parameter M^2 — and holds that the *form* of every structure is derivable from these primitives while the *content* is realized by contingent draw. The monograph presents, in self-contained form: the formal primitives and dynamical system (the graph Laplacian L_Q and the Schrödinger equation); the verified continuum-limit chain from L_Q to the flat Laplacian, the d'Alembertian, the Laplace–Beltrami operator, the curved-space Schrödinger equation, and the Einstein tensor; the derivation hierarchy (gauge structure, spatial dimensionality, the multiplicity bound, the abundance law, phase-gradient gravity); the three-category taxonomy; the conditional no-go theorems; the quantitative predictions; and the complete inventory of executable verification results.

In addition to the technical content of the reference edition, this edition adds an explanatory layer: the historical *origin* of the theory and its transition from its predecessor framework; a *development timeline* recording the major phases, key discoveries, and abandoned routes; an account of *why* the structure/content split is the organizing principle; the *hostile-review history* and what survived it; a program-by-program *research journey*; and the *lessons learned* from wrong assumptions, rejected derivations, and no-go discoveries. Throughout, each major theorem is accompanied by its *intuition*, *significance*, and *limitations*. Nothing in this narrative layer changes the physics or the conclusions of the reference edition.

Publication status: READY AS A FOUNDATION MONOGRAPH — READY AS A RESEARCH-PROGRAM PAPER — NOT READY AS A DERIVATION PAPER.

A foundation monograph may legitimately use imported proven theorems as inputs; the remaining blockers affect only a *derivation-paper* claim. This monograph is released for archival on Zenodo as a foundation monograph and a research-program record; it is *not* submitted as a peer-reviewed derivation paper. The central derivation claim (Einstein recovery from Q-events) remains *logical, not mathematical*: the metric and the BDG action are imported (proven but not TQM-derived), $G = \ell^2 c^3/\hbar$ is dimensional analysis, and no unique sharp prediction yet discriminates TQM from SM+ΛCDM. Every claim marked **verified** is backed by an executable xUnit test in `TQM.Tests/ResearchXC` (or `ResearchQG`); see Part V and Appendix A.

Contents

Genesis and Method

This part records how the theory came to be, why it is organized as it is, and what the research program actually did. It is historical and methodological, not physical: it introduces no new claim and revises none. Its purpose is to make the technical parts that follow legible as the outcome of a specific, auditable research trajectory rather than as a finished axiomatic system dropped from nowhere.

Chapter 1

How to Read This Monograph

This monograph is a reference edition with an explanatory layer, and the two layers can be read independently. The *reference* layer (the technical parts and the appendices) states the theory, its derivations, its classifications, and its verification record. The *narrative* layer (this part) explains how the theory came to be and why it is organized as it is. Nothing in the narrative layer changes the physics; it exists to make the physics legible as the outcome of a specific, auditable program rather than as a finished axiom system dropped from nowhere.

Three conventions govern the whole document and should be read first.

(1) The taxonomy. Every result is tagged with one of three statuses. **DERIVED** means the result is computable from the primitives by a theorem. **REAL-UNDERIVED** ("real-underived") means the structure is real, precise, and predictive, but its origin is not computable from the primitives. **DRAWN** means the value is a coincidental draw of the abundance law. The tags are claims *about derivability*, not about truth: a **REAL-UNDERIVED** or **DRAWN** result can be exactly right while remaining underived. Reading a **REAL-UNDERIVED** tag as a derivation was the single most common misreading the hostile review had to correct.

(2) What "verified" means. A result marked **verified** is backed by an executable xUnit test in the repository. "PASS" means the test ran and its assertion held; it does *not* mean the result was derived from the primitives — that is the taxonomy's job. Some tests verify a derivation; others verify a *standard* mathematical identity (such as the Einstein-tensor chain) precisely in order to show where the TQM derivation stops.

(3) What "confidence" means. The confidence numbers (e.g. T-02 = 0.85) are *uncalibrated ordinal ranks*, not posterior probabilities. They order the program's own degree of belief; they do not measure it. They are never to be read as p-values or as numerical posteriors.

A short map. This part gives the history and method. The *Foundations* part states the primitives and the dynamical system. The *Continuum-Limit* part records the two continuum chains. The *Derivation Hierarchy* part runs dimensionality, gauge, flavor, gravity, and cosmology. The *Classification* part states the taxonomy, the no-go theorems, and the predictions. The *Verification* part gives the test record, the hostile-review audit trail, and the conclusion. The appendices carry the full test inventory, the confidence tables, a worked derivation, the program ledger, the glossary, and the references.

Chapter 2

The Origin of TQM

2.1 Motivation

THE Q-MODEL (TQM) began from a single, deliberately simple question: are matter, quantum behaviour, and gravitation *fundamental*, or do they emerge from something more basic? The working assumption that has guided the entire program is the claim that *matter is not fundamental* — that what we call a particle is a dynamically stabilized wave structure, and that synchronization and resonance are more fundamental than particles themselves. From that seed a research path was fixed early and never reversed:

Temporal Oscillators → Synchronization → Resonance Clusters → Self-Stabilizing Wave Structures → Proto-Particles → Quantum Behaviour → Gravity → Cosmology.

Four standing decisions disciplined the program from the outset, and they explain much of the monograph's structure. *No cosmology first* and *no gravity first* forbade the temptation to tune a model against the most spectacular data before the microscopic foundations were in place. *No particle assumptions* forbade importing the Standard Model's ingredients (gauge groups, generations, masses) as inputs. *Emergence first* required every ingredient to be earned from below, and *physics before interpretation* required the dynamical content to be settled before any ontological gloss was attached. These rules were designed to prevent the theory from being reverse-engineered toward known physics and to force every element to carry its own weight.

2.2 Predecessor ideas

TQM did not appear in a vacuum. Its immediate predecessor was the **Temporal Resonance Model** (TRM), a framework built on phase-lattice machinery that had already assembled a number of striking pieces: a *Time Field* (a phase-gradient picture of gravity), a *Radial Acceleration Relation* (RAR), *Frame Dragging*, a *Memory Channel*, a *Theta Chain*, an $m = 3$ *Closure* mechanism (rational mode-locking $\Omega = (q + 3)/q$, $\gamma = 1/\Omega$), a *Quantum Engine*, a *Temporal Drift* (tired-light) module, and a *Unified Action* roadmap. TRM supplied the conceptual vocabulary — a temporal field, oscillation, phase, resonance — on which the later theory was rebuilt.

Each module deserves a one-line statement of what it actually claimed. The *Time Field* identified gravity with the phase gradient of a temporal field — the idea that, after the transition, became TQM's phase-gradient gravity picture. The *Radial Acceleration Relation* targeted the galaxy-rotation acceleration scale. *Frame Dragging* introduced a vector sector (gravitomagnetism). The *Memory Channel* proposed a history-encoding invariant $\varphi^2|\dot{\mu}|$. The *Theta Chain* was a nonlocal observable-extraction chain $\Theta \to O_5 \to \lambda_\Theta \to g_{\mathrm{obs}}$. The $m = 3$ *Closure* proposed rational mode-locking ($\Omega = (q + 3)/q$, $\gamma = 1/\Omega$) as a mechanism for the "why 3" question. The *Quantum Engine* was a non-unitary quantum module. The *Temporal Drift* was a tired-light (β_T) cosmology. The *Unified Action* was a roadmap $S_{\mathrm{eff}}[T, A_T, \Theta]$ depending on all the others. This list matters because the transition was decided module by module, not wholesale.

The relationship between TRM and TQM is best summarized by the *TRM Legacy Final Classification*, the systematic audit that determined what, if anything, of the older framework should migrate into the new one. Of the nine TRM modules, *three* were *absorbed*: the Time Field became the phase-gradient gravity picture, the RAR became the zero-parameter relation $g_\dagger = cH_0/2\pi$, and Frame Dragging was recognized as GR gravitomagnetism (Lense–Thirring, already measured). *Two* were *rejected*: Temporal Drift (tired light) was falsified by the QG-080–089 reinterpretation audits, and the Quantum Engine was found to be non-unitary and worse than the BDG lattice operator. *Three* survived as *candidate mathematics only*: the $m = 3$ Closure (a candidate mechanism for the $N \leq 3$ upper bound, but unmapped to observables), the Memory Channel (a genuine invariant $\varphi^2|\dot{\mu}|$ with no observable), and the Theta Chain (a homonym of TQM's own information field). *One* remained open: the Unified Action roadmap.

2.3 The TRM → TQM transition

The decisive result of the legacy audit was negative and, in its way, liberating: TRM carried *zero candidate physics* — no TRM module supplied a genuinely new testable observable. The residue of value was methodological. What the transition produced was therefore a *clean reconstruction* rather than an extension: the new theory re-derived, from the fewest possible ingredients, exactly the structure that TRM had assembled piecemeal, and discarded the parts that did not survive hostile reduction.

The compressed endpoint of that reconstruction is the subject of this monograph: a theory with *two irreducible primitives* — the individuation principle Q and genuine ontological randomness — plus *one* contingent continuous number M^2 (the nonlinearity regime), with everything else — time, space, 3+1 dimensions, Hilbert space, unitary quantum mechanics, the Schrödinger equation, the Born rule, measurement, gravity as causal structure, particles as topological defects, and $U(1)$ gauge symmetry — derived or emergent. The parameter count collapses from the Standard Model's ~ 19 free numbers to 2 primitives plus 1 number, a reduction of roughly 95%, while $U(1)$ is promoted from an assumption to a theorem.

Chapter 3

Development Timeline

The program is best read as a sequence of phases, each of which closed one question and opened the next. This chapter gives the skeleton; the detailed experiment records live in the project's laboratory book, and the four research programs are narrated individually in Chapter 6.

3.1 Major phases

The program falls into five successive eras. Each is summarized here in enough detail to show what was actually done; the four research programs that carry the technical weight are narrated further in Chapter 6.

3.1.1 Early oscillator experiments (TQM-001–TQM-008)

The first experiments were numerical and exploratory, and their role was to find out whether the central intuition — coherent structure emerging from oscillators — had any teeth. TQM-001 showed that an order parameter R of a population climbs toward 1: coherent synchronization is not put in by hand, it appears. TQM-002 then closed a tempting blind alley: random matrices give Wigner-like spectra but do *not* produce the kind of structured emergence the program needed, so randomness alone is not enough. TQM-003 showed that structured topology alone is also insufficient — the dynamics, not the static graph, carries the weight. TQM-004 showed that a field-mediated interaction does not by itself synchronize two asymmetric oscillators; the field behaves more like a resonance medium than a pure synchronization medium.

TQM-005 found stable resonance clusters of three oscillators forming and persisting over most of a simulation — early, tentative support for "matter as stabilized resonance" — but with only a few percent energy participation, so the clusters did not yet dominate the field. TQM-006 found the first sharp threshold, a critical resonance density $\rho_c \approx 0.09$, above which persistent coherent modes appear: a percolation-like transition, and the program's first well-defined critical point. TQM-007 reported five resonance families. TQM-008 then demolished that apparent diversity with a 400-simulation reproducibility study, leaving a single universal coherent mode (F4) and exposing the others as seed artifacts. The lesson of TQM-008 — apparent diversity can be a single seed — became a standing methodological caution for everything that followed.

3.1.2 Alternative foundations (ResearchX, X001–X034)

A parallel track attacked the foundations from the side, and it is responsible for several of the theory's deepest results. X001 found eleven hidden assumptions, with the static graph chief among them. X002–X004 tested dynamism (node motion, mobility, graph growth) and found no new phenomena: dynamic graphs alone produce no open-ended innovation. X005– X006 showed that nonlinearity breaks eigenmodes and creates solitons, so the "quantum" reading of the early linear results had to be withdrawn — Q, fitness, and evolution survived, the naive quantum reading did not. X007–X015 built the general theory of information carriers and proved that *reversibility* plus *self-consistency* is the minimally sufficient pair whose intersection is unitary quantum mechanics.

X016–X022 built the complexity staircase and proved that open-ended evolution cannot occur in a finite system (a pigeonhole bound). X023–X031 then proved that quantum reality is the unique global maximum of finite complexity — a necessity result, not an anthropic one. X032–X034 unified the main line with ResearchX, confirming that L_Q is not fundamental: any operator satisfying reversibility and self-consistency at their joint maximum yields unitary quantum mechanics. This is the result behind the claim that the theory has been compressed to its minimal form.

3.1.3 Spacetime emergence (X040–X046)

Time was derived as the partial order of actualization events ($E_1 < E_2$ iff E_2 depends on E_1's outcome), with the arrow of time inherited from the irreversibility of actualization. Gravity was placed on a causal-set footing, and a reconstruction experiment recovered distances from Q-event correlations. Dimensionality 3+1 was derived from complexity maximization (capacity times stability times accuracy). Newton's constant, $\hbar$, and Λ were all traced to the single event scale ℓ: $G = \ell^2 c^3/\hbar$, $\hbar = 1$ in event units, and $\Lambda \sim 1/\sqrt{N}$ from Poisson fluctuations of the event count. This is the era in which the theory stopped needing to *assume* spacetime and began to *produce* it.

3.1.4 Matter emergence (X047–X060)

Particles were identified with topological defects (solitons): charge with the Betti number, mass with defect energy, spin with orientation. $U(1)$ was derived as the automorphism group of the circle. Generations were characterized as excitation levels of the same topology, with the mass hierarchy following an anharmonic spacing. Fermion mixing was traced to defect wavefunction overlap (hierarchical CKM from Dirac defects, anarchic PMNS from Majorana defects). Neutrino masses were explained by delocalized defects, and normal ordering was predicted from an attractive self-interaction. The Standard Model gauge group was shown to be *preferred* by ecological fitness but *not* uniquely derivable — the largest remaining gap in the matter program.

3.1.5 Parameter compression and closure (Phases 140–159)

The final era consolidated. The parameter-count audits (X057–X060f) reduced the theory to two primitives plus one number, with $U(1)$ promoted to a theorem and M^2 identified as the single irreducible continuous parameter. The late phases then closed the residual questions one by one: the RAR derivation $g_\dagger = cH_0/2\pi$; the Koide closure (seven routes exhausted, six no-gos plus one blocked); the gauge-origin audit ($U(1)$ derived, $SU(2)$ emergent, $SU(3)$ contingent); the multiplicity and $N \leq 3$ audits; the internal-3 disposition; the cascade-unification audit; the hostile-review response; and the v1.0 release. Each of these closures is narrated where it appears later in the technical parts.

3.2 Key discoveries

- Synchronization and coherent oscillation emerge spontaneously above a critical coupling density.
- Reversibility $\cap$ self-consistency = quantum mechanics: the Schrödinger equation is the unique maximal-complexity dynamics.
- $L_Q \to -\nabla^2$ (the elliptic chain) and BDG $\to \Box$ (the Lorentzian chain) are each controlled continuum limits, but they are *disjoint* in signature.
- $U(1)$ is a theorem ($\mathrm{Aut}(S^1) = U(1)$, $\pi_1(S^1) = \mathbb{Z}$), the cleanest result of the gauge program.
- The Koide relation is real, predictive, and RG-stable — and *underivable* (no-go T-08). The neutrino-Koide analogue is *falsified* (T-11).
- The RAR $g_\dagger = cH_0/2\pi$ matches a_0 with zero free parameters.
- Complexity maximization selects $d = 3+1$.

3.3 Failed routes

A research program is defined as much by what it abandoned as by what it kept. The principal abandoned routes were: random matrices as an emergence mechanism; static topology as a sufficient condition; the apparent resonance-family diversity of TQM-007 (artifacts of a single seed); dynamic graphs as a source of open-ended innovation; the identification of solitons as *quantum* objects (they are a nonlinear generalization of carriers, not the quantum regime); open-ended evolution in finite systems (proved to be delayed saturation); symmetry, attractor, topology, and information-geometry routes to the Koide relation; topology, attractor, moduli, and persistence routes to the gauge count; stability, anomaly, representation, defect, and information routes to $N \leq 3$; tired-light cosmology; and the TRM Quantum Engine. Each failure was recorded as a no-go or a negative result, and most were later promoted to the no-go theorems of Chapter 23.

Chapter 4

Why the Structure/Content Split

4.1 Historical motivation

The structure/content split — *structure is derivable; content is realized* — was not an axiom chosen in advance. It was the *conclusion* of a specific audit (QG-042, the "Parameter vs. Structure" study) that asked a blunt question: given the primitives, what can actually be *computed*, and what merely *is*? The answer divided cleanly along a line nobody had drawn explicitly before.

Everything that could be derived — oscillation, phase, $U(1)$, charge quantization, gravity, inertia, the equivalence principle, particle stability, the interpretation of the Higgs field — turned out to be *form*, *symmetry*, or *topology*. They answer *what exists* and *why*. Everything that resisted derivation — $\alpha = 1/137$, α_s, θ_W, the Yukawa couplings, λ, θ_{QCD} — turned out to be *dimensionless pure numbers*. They answer *how much*. The deep divide maps exactly onto the two primitives: Q underwrites structure (laws), and Random Actualization underwrites content (history).

There is a sharp reason the divide is *fundamental* rather than temporary. No combination of the dimensionful triple $(\ell, \tau, \hbar)$ can produce a dimensionless number, so a dimensionless coupling *cannot* be assembled from the unit-convention primitives. The theory therefore does not merely *fail* to derive the couplings; it explains *why they cannot be derived*: they are historical outcomes of the actualization draw, not mathematical necessities. Contingency is intrinsic; no multiverse is needed. Structure is what randomness cannot change; content is what it determines.

4.2 What the split is not

Three confusions must be headed off, because the hostile review raised all three. First, the split is *not an immunization strategy*: it does not say "whatever we cannot derive is contingent". The boundary is not moved after the fact; it is a fixed, testable division — structure is what the primitives compute, content is what they cannot — and the theory accepts the consequences, including the underivability of its own couplings. Second, the split is *not a multiverse*: contingency is intrinsic to a single universe through Random Actualization, and no ensemble of universes is invoked. Third, the split is *not anthropic*: the derived pieces (spatial 3, $U(1)$, the log-normal form) are theorems or distribution-shape results, not survival-condition selections, and the selected pieces are flagged as selected rather than hidden behind the word "derived".

The point of the split is precisely to make the theory *falsifiable in its classifications*. A result tagged **REAL-UNDERIVED** is a standing invitation to find the derivation that was missed; a result tagged **DRAWN** is a standing prediction that the value has no deeper origin. Both are testable claims about the boundary of derivability, not rhetorical immunizations.

4.3 Examples

The split is easiest to grasp through concrete cases, each of which appears in full later:

- **Gauge structure.** $U(1)$ (the group) is **DERIVED**; the color count $n = 3$ (the content) is **DRAWN**. The group structure is computable from $\mathrm{Aut}(S^1)$; the count is not fixed by any tested principle (T-09).
- **Dimensionality.** Spatial 3 is **DERIVED**(complexity maximization selects $d = 3+1$); internal 3 (generations, color) is *selected* — a derived lower bound $N \geq 3$ meets an empirical upper bound $N \leq 3$.
- **The abundance law.** The log-normal *form* of the abundance distribution is **DERIVED**(a multiplicative cascade plus the central limit theorem in log-space); the realized values μ, σ are contingent content.
- **The Koide relation.** $Q = 2/3$ is real, predictive, and RG-stable — a canonical **REAL-UNDERIVED**structure — but its origin is underivable (T-08).
- **Yukawas.** The overlap-operator *form* of the Yukawa operator is derived; its *spectrum* (the hierarchy 1:207:3478) is set by the contingent architecture shapes.

This taxonomy of underivability — **DERIVED**, **REAL-UNDERIVED**, **DRAWN**— is formalized in Chapter 22. Its function is honesty: it says precisely which claims are theorems and which are classifications, and it is the reason the hostile-review history (Chapter 27) is a history of *framing* corrections rather than of retracted physics.

Chapter 5

Hostile Review History

5.1 Initial criticisms

The theory was subjected to two rounds of hostile review, and the record of both is part of the project. The first review (of the v1.0 paper) raised twelve issues; the response audit classified them as *1 valid*, *11 partially valid*, and *0 invalid*. The dominant failure mode was not a fatal theory flaw but a *documentation* gap: the paper omitted the dynamical system, the formal primitives, the complexity functional, the emergent-GR matching, and the quantitative predictions that the program actually contained, and it overstated its framing ("closed" where it should have said "dispositioned"; "no-go" where it should have said "conditional no-go"; "derivation" where it should have said "ontological reinterpretation"). Three genuine theory gaps were acknowledged: the provisional gauge-count no-go (T-09), the contingent content, and the immunization risk of the structure/content split.

The response audit also identified seven release blockers, all of them documentation fixes with no new physics: six missing sections (the formal primitive definitions, the dynamical-system summary, the complexity functional, the emergent-GR derivation summary, the quantitative predictions, and the scope-and-limitations statement) plus three wording fixes that were treated as a single framing correction. The important point is the classification: the review found *one* genuinely valid objection (the "no route open" framing), *eleven* partially valid objections that pointed at omitted documentation rather than at a wrong result, and *zero* fully invalid objections. A review that lands in that distribution is a review of a paper's *presentation*, not of its *core* — and the program treated it accordingly: fix the presentation, keep the core, and re-verify nothing changed.

A second round (Round 2) raised twelve "FATAL" issues, tabulated in Chapter 27. They ranged from "primitives undefined as mathematical objects" to "no controlled continuum limit."

5.2 What changed

The response was a revision that corrected every framing overstatement and added the six missing sections (formal primitive definitions, dynamical-system summary, complexity functional, emergent-GR derivation summary, quantitative predictions, scope and limitations). The wording fixes were: "closed" → "dispositioned", "no-go" → "conditional no-go", and "derivation" → "ontological reinterpretation" where appropriate. No physics changed.

5.3 What survived

The one substantive objection that could not be disposed of by documentation was "no controlled continuum limit." It was met *fully* on the Schrödinger side ($L_Q \to -\nabla^2 \to$ Schrödinger, plus the curved L_W, the Laplace–Beltrami approximation, and unitarity, all now tested) and *partially* on the Einstein side (the standard chain works, but the metric and the BDG action are imported). The decisive residual is therefore the *native metric → operator coupling* — the "G4" gap: TQM imports $g_{\mu\nu}$ via Malament's theorem rather than generating it from Q-events. That residual is what keeps the verdict at **READY AS A FOUNDATION MONOGRAPH**

— READY AS A RESEARCH-PROGRAM PAPER — NOT READY AS A DERIVATION PAPER.

Chapter 6

The Research Journey

The derivation program was organized into research programs, each closed by executable experiments. Four of them carry the technical weight of this monograph and are narrated here.

6.1 The gauge program

The gauge program asked why the gauge group is $SU(3) \times SU(2) \times U(1)$. The answer is tripartite, and each part has a different status.

$U(1)$ **is derived.** Phase lives on the circle S^1; its isometry group *is* $U(1)$, and the winding $\oint \nabla\theta \cdot dl = 2\pi n$ yields integer charge, with the gauge connection $A_\mu = \partial_\mu \theta$ and the photon as the $n = 0$ phase wave. This is the cleanest result of the entire program: not a choice but the symmetry of phase itself.

$SU(2)$ **is emergent.** The binary winding pair $\{n = \pm 1\}$ is the minimal doublet; the Bloch sphere gives $SO(3) = SU(2)/Z_2$; the spinor double cover gives $SU(2)$. The "2" is near-derived, but the lift from the discrete Z_2 to the continuous $SU(2)$ is not itself derived — it is the one admitted gap in the $SU(2)$ story.

$SU(3)$ **is contingent.** The $\mathrm{Aut}(C^n/S_n) \supseteq SU(n)$ argument derives the group *structure* but not the *count*, and the 8-gluon algebra is borrowed. Four route families were tested for the count: topology alone (fails for $n > 1$, since $\pi_1(S^1) = \mathbb{Z}$ is infinite), attractor space (fails, contingent), defect moduli (the strongest, but it derives structure, not count), and persistence/symmetry (fails to select). The 1–2–3 rank-equals-winding pattern is intriguing but "possibly numerology": no mechanism links the three ranks. The result is no-go T-09 — no topological, attractor, moduli, or persistence principle fixes the defect count — and the residual "why 3" is the theory's one open door.

6.1.1 Failed paths and why they failed

Four route families failed for the defect count. *Topology alone* fails because $\pi_1(S^1) = \mathbb{Z}$ is infinite and cannot select $n = 3$. *Attractor space* fails because the landscape content is contingent, so it cannot select a count. *Persistence and symmetry* fail because every classical group is stable, so stability does not discriminate. The *rank-equals-winding* "1–2–3" pattern is intriguing but has no linking mechanism — "possibly numerology". The defect-moduli route was *not* a failure: it is the strongest route, but it derives the structure $\mathrm{Aut}(C^n/S_n) \supseteq SU(n)$ and stops short of the count, which is exactly the boundary the no-go (T-09) marks.

6.2 The Koide program

The Koide relation $Q = \sum m/(\sum \sqrt{m})^2 = 2/3$ is real to $\sim 10^{-5}$, predictive (1981 prediction $m_\tau = 1776.97\,\mathrm{MeV}$, confirmed 1992), and RG-stable. The Koide program asked where it comes from, and answered by *exhaustion*: it tried every route it could think of and recorded which failed.

Seven routes were exhausted — symmetry (S_3), topology (S^1), attractors, information geometry, group theory, the $m = 3$ closure, and moduli arguments — with the result *six no-gos and one blocked.* Four computed origin tests all failed: S_3 democratic mixing gives $Q = 1/3$ (not 2/3); there is no RG fixed point at 2/3; S^1 locates but does not fix the angle; and the information-geometry measure $S/S_{\max} = 0.5093$ is not extremal.

Five fallacies were explicitly rejected: anthropics ($Q = 2/3$ is not anthropic), texture fitting (fits, does not derive), numerology (real but not derived), hidden parameters (forbidden), and 45° restatements (they add zero information).

The closure is a *no-go* (T-08): the Koide origin is a closed question — real, predictive, RG-stable, *underivable* — the canonical **REAL-UNDERIVED**structure. The neutrino-Koide analogue was then tested and *falsified* (T-11: $Q_{\max} = 0.585$ normal / 0.500 inverted, both $< 2/3$), removing the last remaining distinguisher and closing the program.

6.2.1 Failed paths and why they failed

Seven routes failed, and the pattern of failure is informative. *Symmetry (S_3)* fails because democratic mixing gives $Q = 1/3$ and a hierarchical limit gives $Q = 1$; 2/3 is the non-generic midpoint. *Topology (S^1)* fails because it locates the phase without fixing the 45° angle. *Attractors* fail because there is no RG fixed point at 2/3. *Information geometry* fails because $S/S_{\max} = 0.5093$ is not extremal. *Group theory* fails to select. The $m = 3$ *closure* fails because its $\gamma \neq 2/3$ and its parameters are unmapped to observables. *Moduli* is blocked, not no-go. The rejected fallacies — anthropics, texture fitting, numerology, hidden parameters, 45° restatements — failed because each either fits without deriving or restates without adding information.

6.3 The continuum program

The continuum program asked whether the discrete dynamical system has a controlled continuum limit, and it split into two halves with unequal outcomes.

On the Schrödinger side the answer is an unambiguous *yes*. The chain Laplacian L_Q converges to the flat Laplacian $-\nabla^2$ with an exact closed-form spectrum and second-order convergence, and the weighted Laplacian L_W supplies a curved generalization and a Laplace–Beltrami approximation. This half is fully tested.

On the Einstein side the answer is more guarded. The BDG causal-set operator converges to the d'Alembertian $\Box$ with $O(h^2)$ control, but L_Q and BDG are *disjoint in signature* — one elliptic and positive, the other hyperbolic and indefinite — so no native bridge between them has been closed. A dedicated three-test program made the incompatibility executable: L_Q is positive semi-definite, BDG is indefinite, and the graph Laplacian is explicitly rejected for Lorentzian signature. This — two disjoint chains on the same substrate — is the single most important structural fact of the program, and it is the precise reason the Einstein recovery is logical rather than mathematical.

6.3.1 Failed paths and why they failed

The one path that failed — and the reason it failed — is the naive hope that a single operator could serve both the Schrödinger and Einstein sides. It failed because the two continuum limits have incompatible signatures: $L_Q \to -\nabla^2$ is elliptic and positive semi-definite, while BDG $\to \Box$ is hyperbolic and indefinite. The executable test made the failure precise: the graph Laplacian is explicitly rejected for Lorentzian signature. The program therefore did not produce a missing "bridge"; it produced a *no-bridge result* — two disjoint chains on the same substrate — which is the most important structural fact of the program.

6.4 The metric program

The metric program asked how a metric $g_{\mu\nu}$ can arise at all. The answer is a four-link chain: Q-events $\to$ causal order (native) $\to$ conformal class (proven, via Malament's imported theorem) $\to$ conformal factor (native, from the counting measure $f = \rho^{2/d}$) $\to$ the metric. The metric *origin* is therefore closed: the only imported link is a proven theorem, not a theory gap. The verification is direct — the causal order satisfies the metric axioms, the null structure is invariant under conformal rescaling but not under non-conformal rescaling, and the counting measure round-trips the conformal factor.

What remains open is the metric *dynamics* — the native metric $\to$ operator coupling (G4). Closing the origin tells us *where* the metric comes from; it does not tell the operator how to *use* it. That residual is why the Einstein recovery remains logical rather than mathematical, and it is the single genuinely-open technical item in the monograph.

6.4.1 Failed paths and why they failed

The path that failed is the *native* metric dynamics. TQM does not produce $g_{\mu\nu}$ from Q-events; the metric-dependent operator $\Delta_g/\Box_g$ is absent, and the weighted Laplacian L_W supplies only an approximation whose defining metric is external. The attempt to close the metric *dynamics* natively therefore failed, and the program's honest endpoint is the split it records: the metric *origin* is closed (Q-events $\to$ causal order $\to$ conformal class $\to$ conformal factor $\to$ metric, with Malament's theorem imported as a proven link), while the metric *dynamics* remain imported. That split is the "G4" gap.

6.5 What the four programs establish together

Read together, the four programs trace one arc. The gauge and Koide programs locate the *boundary* between the derivable and the contingent; the continuum and metric programs locate the *boundary* between the controlled and the imported. Both boundaries are honest: the theory claims to derive structure and to control the Schrödinger limit, and it explicitly declines to claim the gauge count, the Koide origin, or the native Einstein bridge. That division — derivable structure, controlled Schrödinger limit, classified content, imported metric — is the shape of the finished theory.

Chapter 7

Lessons Learned

7.1 Wrong assumptions

- **The static graph was the number-one hidden assumption.** The first alternative-foundations audit (X001) found eleven hidden assumptions, with the static graph first on the list. When it was tested (X002–X004), dynamic graphs produced no new phenomena and no open-ended innovation.

- **Nonlinearity breaks the eigenmode picture.** X005 showed that most of the early "quantum" results were linear-algebra artifacts: nonlinearity destroys eigenmodes and creates solitons instead. Q, fitness, and evolution survived; the naive quantum reading did not.

- **Open-ended evolution in finite systems is an illusion.** X025 appeared to show open-ended species growth, but X026–X027 proved it was delayed saturation: a pigeonhole argument bounds the number of species by the number of vertices, so a finite system always saturates. Level-6 (open-ended) evolution requires an infinite state space.

- **Apparent diversity can be a single seed.** The TQM-007 resonance "families" collapsed to one universal mode (F4) under a 400-simulation reproducibility study; the rest were artifacts.

7.2 Rejected derivations

The program repeatedly attempted — and rejected — candidate derivations rather than settling for postdiction: symmetry, attractor, topology, and information-geometry routes to Koide (T-08); topology, attractor, moduli, and persistence routes to the gauge count (T-09); stability, anomaly, representation, defect, and information routes to $N \leq 3$ (T-10); five separate attempts to derive $\alpha = 1/137$; tired-light cosmology; the TRM Quantum Engine; and the grand-unified groups $SU(5)/SO(10)$ (proton decay unobserved) and E_6/E_8 (exotic particles unobserved). In each case the rejection was recorded, and where the failure was systematic it became a no-go theorem.

7.3 No-go discoveries

The no-go theorems (Chapter 23) are the program's most distinctive scientific product. T-08 (Koide underivable), T-09 (gauge count unfixed), T-10 ($N \leq 3$ unfixed), T-11 (neutrino-Koide falsified), and T-12 (shared cascade untestable) collectively define the boundary between what the theory can compute and what it must classify. The meta-lesson is the structure/content split itself: the no-gos are not failures of ingenuity but the expected signature of a theory in which content is contingent by construction.

7.4 Meta-lessons

Beyond the specific wrong assumptions and rejected derivations, the program yielded a small number of standing lessons that shaped everything after them.

The derivability boundary is itself a result. The discovery that structure is derivable while dimensionless content is not (QG-042) is not a limitation the program stumbled into; it is a positive finding, and it is what licenses the taxonomy. A theory that *explains why* its couplings cannot be derived is making a stronger claim than a theory that merely fails to derive them.

Negative results are first-class products. The no-go theorems (T-08–T-12) are the program's most distinctive scientific output, not its embarrassments. A route that is shown to be closed is information, and recording it as a theorem (rather than quietly dropping it) is what makes the theory auditable.

Honesty in framing is a scientific asset. The hostile-review history is almost entirely a history of framing corrections, not retracted physics. The lesson is that "closed" vs. "dispositioned", "no-go" vs. "conditional no-go", and "derivation" vs. "ontological reinterpretation" are not cosmetic distinctions; they are the difference between an overclaim and a defensible statement.

Executable verification is what makes honesty checkable. Every "PASS" in this monograph is the output of a test that a hostile reviewer can run. That discipline is what turned the hostile-review objections into an audit trail that could be answered point by point rather than argued about indefinitely.

Chapter 8

Evolution of the Framework

The framework did not arrive at its final form in one step; it was repeatedly stripped down and rebuilt. This chapter traces what was assumed at the start, what was removed, what was retained, and the two methodological commitments that emerged — the structure/content split and the no-go method.

8.1 Earliest assumptions

The program began with a rich, and mostly wrong, set of assumptions. Matter was assumed to be a dynamically stabilized wave structure in a temporal field; synchronization and resonance were taken to be more fundamental than particles; and the route from oscillators to cosmology was assumed to be a single continuous chain. Implicitly, the early experiments also assumed that a static graph was the right substrate, that random matrices might be sufficient for emergence, and that whatever coherent structure appeared would carry the quantum reading of the system.

8.2 Assumptions removed

Most of the earliest assumptions were removed by experiment. Random matrices did not produce emergence (TQM-002); static topology alone was insufficient (TQM-003); the apparent diversity of resonance families collapsed to a single reproducible mode (TQM-008); dynamic graphs produced no open-ended innovation (X002–X004); and nonlinearity showed that the early "quantum" results were linear-algebra artifacts (X005). The specific assumptions of the predecessor TRM framework — tired-light cosmology and a non-unitary quantum engine — were rejected outright in the legacy audit. What remained after this stripping was minimal: individuation (Q), reversibility, self-consistency, and genuine randomness.

8.3 Assumptions retained

Two pairs survived every hostile reduction. *Reversibility* (conserved norm) and *self-consistency* (fixed points) were proved to be independent and minimally sufficient: their intersection is unitary quantum mechanics (X011–X015). Q (individuation) and *genuine randomness* were shown to be irreducible — no deeper invariant exists below them (X035, X039). Everything else in the theory — time, space, 3+1, the Schrödinger equation, the Born rule, measurement, gravity as causal structure, particles as defects, and $U(1)$ — is derived or emergent from these, plus the single contingent number M^2.

8.4 Emergence of the structure/content split

The structure/content split was not imposed; it was *discovered*. The parameter-vs- structure audit (QG-042) found that everything derivable from the primitives was *form*, *symmetry*, or *topology*, while everything that resisted derivation was a *dimensionless pure number*. Since no combination of the dimensionful triple $(\ell, \tau, \hbar)$ can produce a dimensionless number, the couplings *cannot* be assembled from the primitives. That turned a temporary limitation into a principle: structure is derivable, content is realized, and the taxonomy (**DERIVED**, **REAL-UNDERIVED**, **DRAWN**) is the honest accounting of that boundary.

8.5 Emergence of the no-go methodology

The no-go method emerged from the same discipline. When a derivation failed — Koide, the gauge count, $N \leq 3$ — the program did not discard the failure; it enumerated every route it could think of, tested each against the no-new-primitives constraint, and recorded the result as a conditional no-go theorem (T-08–T-12). What began as a way to close individual questions became a general method: *route exhaustion* plus *hostile reduction*, which turns "we could not derive it" into the stronger, checkable statement "no tested route derives it."

Key results. The framework was stripped from a rich oscillator picture to two primitives plus one number; the structure/content split and the no-go method emerged as its two organizing commitments.
Open questions. Whether a native metric-to-operator coupling (G4), a unique discriminating prediction, or a native BDG action can be derived — the three derivation-paper blockers.
Verification status. Historical/editorial; no executable test applies, but every removal and retention cited here is recorded in the development ledger (Appendix).

Chapter 9

The Method of Executable Audits

The single most distinctive feature of this program is not a specific result but a *method*: every substantive claim is made executable, and every "-audit" in the record is a piece of that method. This chapter explains the machinery, because the confidence statements scattered through the technical parts are unintelligible without it.

9.1 What an audit is

An audit is a structured, reproducible investigation with a single question and a disposition at the end. It is implemented as an analyzer (a small program) plus one or more tests, and it writes a report. The question is stated first, the routes are enumerated, and each route is tested against a fixed set of constraints — typically the *no-new-primitives* constraint, and a rejection of anthropics, numerology, hidden dimensions, and post-selection unless they are explicitly named. The disposition is one of a small vocabulary: **DERIVED**, **REAL-UNDERIVED**, **DRAWN**, **no-go**, **blocked**, **contingent**, or **selected**. Because the report is written to a file and the test asserts on the computed numbers, a later reviewer can rerun the audit and get the same answer.

9.2 Route exhaustion and the no-go method

The no-go theorems are produced by *route exhaustion*: enumerate every derivation route the program can think of for a target, test each, and record which fail. A no-go is the statement "within the tested route space and the no-new-primitives constraint, no route reaches the target." It is therefore *conditional* — it is not a proof of logical impossibility (except where a computation, such as T-11, closes the question directly), and it is only as strong as the enumerated route space. The Koide closure is the paradigm: seven routes, six no-gos, one blocked, zero open, and the conclusion is a *closed question*, not a derivation.

9.3 Confidence ranks

The confidence numbers are *uncalibrated ordinal ranks*. They are assigned by the audit to order the program's own degree of belief in a disposition, and they deliberately carry no statistical meaning. A theorem with a complete route-exhaustion behind it (such as $U(1)$, confidence 0.95) ranks above an emergent structure with an admitted gap ($SU(2)$, 0.70), which ranks above a provisional no-go ($SU(3)$ structure, 0.10). The number is a summary of *how much of the route space has been closed*, not a posterior probability.

9.4 Hostile reduction as a method

The program treats hostile review not as an external event but as a standing internal method: each result is periodically subjected to a *hostile reduction* that tries to derive it away, explain it away, or show it is a restatement. The surviving results are exactly the ones that hostile reduction cannot remove. This is why the hostile-review history (Chapter 5) is so short on retracted physics: the framing was corrected, but the underlying results had already been through hostile reduction internally before the external referee saw them.

Part I — Foundations

Chapter 10

The Primitives

This chapter states the theory. Everything that follows in the technical parts is an unfolding of these definitions: the primitives are the whole of the input, and the monograph's burden is to show how much of observable structure follows from them and to say honestly where it stops.

10.1 Statement of the theory

THE Q-MODEL rests on a single organizational claim, the **structure/content split**:

Structure is derivable; content is realized.

The *form* of observable structure (topology, symmetry, distribution shape, lower bounds) is provable from a minimal primitive set. The *content* (specific masses, couplings, multiplicities, angles) is a draw from a contingent ensemble and is therefore classified rather than derived.

10.2 The four primitives

Definition 10.1 (Primitive set)**.** *The theory admits exactly four primitives, and no others:*

1. *Q — the principle of individuation (ontology): the irreducible act by which a discrete event comes to be individuated from nothing.*
2. ***Random Actualization** — genuine ontological chance: given Q, the realized event locations are random within the causal structure. (Assumption **A-03**.)*
3. *$(\ell, \tau, \hbar)$ — the irreducible physical triple: a spacetime scale ℓ, a clock τ, and an action $\hbar$ (unit conventions).*
4. *M^2 — the single continuous nonlinearity parameter (the nonlinearity regime of the dynamics).*

No gauge-group, multiplicity, or hidden-dimension primitive is permitted. The primitives have two layers that meet at Q: an *ontology* layer (underwriting the derivation of structure) and a *dynamical* layer (underwriting the dynamical system of Chapter 12).

Remark 10.2. *Q individuates; Random Actualization realizes; $(\ell, \tau, \hbar)$ fixes units; M^2 sets the single continuous parameter that the derivation hierarchy pins to $M^2 \approx 5$ (Chapter 17).*

10.3 The quantum postulates (dynamical layer)

Axiom 10.3 (Q exists)**.** *Topological charge quanta have position x_i and phase θ_i, and interact pairwise with coupling $J_{ij} = f(|x_i - x_j|)$. (Assumed, irreducible.)*

Axiom 10.4 (Reversible dynamics)**.** *The state norm* $\|\psi\|^2$ *is conserved; equivalently, evolution is unitary.*

Axiom 10.5 (Born rule)**.** $P = |\langle\phi|\psi\rangle|^2$*, uniquely selected by additivity (Gleason's theorem, 1957).*

Axiom 10.6 (Measurement)**.** *A measurement yields one definite outcome (the collapse axiom).*

Postulates 1–2 derive the Hilbert space and the Schrödinger equation (Chapter 12); postulates 3–4 close the interpretation.

Key results. The four primitives (Q, Random Actualization, $(\ell, \tau, \hbar)$, M^2) and the structure/content split are stated; no further primitive is permitted.
Open questions. The measure and the action functional for Q remain postulates, not derived objects.
Verification status. No executable test applies to axioms; the formalization audits in the next chapter record the object/state/operation status.

Chapter 11

Formalization of Q and Random Actualization

This chapter records the formal status of the primitives, as established by the formalization audits (executable; see Part V). Its purpose is to say, precisely, how much of each primitive is already a mathematical object and how much remains a postulate — the honest boundary that the hostile review pressed on.

11.1 Status of Q

Component	Status	Content
Object	**Formalized**	a discrete event (individuated entity)
State space	**Formalized**	positions x_i and phases θ_i
Operations	**Formalized**	individuation, coupling J_{ij}
Configuration space	**Partial**	graph of events, no continuum limit native
Dynamics	**Partial**	L_Q generator (postulate 1)
Symmetries	**Partial**	S^1 phase, Z_2, permutation S_n
Continuum limit	**Partial**	verified for L_Q (Part II)
Measure	**Missing**	no measure space from primitives alone

Classification: Q is *partially formalized* — object, state space, and operations exist; the measure and the action functional remain as postulates.

11.2 Status of Random Actualization

Random Actualization is **A-03**, an assumption, not a derived object. The formalization audit found:

Component	Status	Content
Random variable	**Formalized**	the draw of event locations
Generator	**Formalized**	ontological chance
$(\Omega, \mathcal{F}, P)$	**Partial**	ensemble unspecified
Ensemble measure	**Partial**	no explicit measure

Classification: the random variable and generator are formalized; the probability space $(\Omega, \mathcal{F}, P)$ and the ensemble measure are only partially formalized.

Remark 11.1. *The hostile reviewer's charge that "Random Actualization is verbalism" is therefore* partially valid*: it is an* admitted *primitive (A-03), not a derived object, and the paper states this explicitly.*

Chapter 12

The Dynamical System

This chapter turns the primitives into a dynamics. The single most consequential move is the identification of the graph Laplacian L_Q as the generator, because its eigenbasis *is* the Hilbert space of the theory and its continuum limit carries the whole Schrödinger program.

12.1 The graph Laplacian L_Q

Definition 12.1 (Graph Laplacian)**.** *For the Q-interaction graph with adjacency A and degree matrix D,*

$$L_Q = D - A. \tag{12.1}$$

Intuition. *Each row sums to zero, so the uniform state is the zero mode and the eigenvectors are the standing modes of the graph.* ***Meaning.*** *It is the discrete diffusion/hopping operator of the theory.* ***Why it matters.*** *Its eigenbasis* is *the Hilbert space, and its continuum limit is the Laplacian that carries the whole Schrödinger program.* ***Limitations.*** *Undirected and unweighted here; the Lorentzian (BDG) and curved (L_W) cases are separate operators.*

L_Q is real, symmetric, positive semi-definite, and has zero row sums. Its eigenvectors form an orthonormal basis — the ***Hilbert space*** *of the theory.*

12.2 The tight-binding identity (TQM-142)

Proposition 12.2 (Tight-binding identity)**.** *On a 1D chain,*

$$(L_Q)_{ij} = 2\delta_{ij} - \delta_{i,j+1} - \delta_{i,j-1}, \tag{12.2}$$

which is identical *to the tight-binding Hamiltonian with on-site energy $\varepsilon = 2t$: $H = tL_Q$. This is a mathematical* ***identity****, not an analogy.*

Intuition. On a chain the Laplacian is exactly the second-difference operator, so "graph diffusion" and "tight-binding hopping" are the same object written twice.

Significance. The identity is what licenses the transfer of solid-state techniques — spectra, Green's functions, band structure — to the ontology layer without any translation.

Limitations. It holds in one dimension and for the bare (unweighted) Laplacian; the curved and causal-set generalizations are separate objects (Sections 14.2 and 13.3).

12.3 Schrödinger from reversibility (TQM-149–151)

Theorem 12.3 (Schrödinger equation from reversibility)**.** *Consider linear evolution $\partial_t\psi = M\psi$ with conserved norm $\partial_t\|\psi\|^2 = \psi^\dagger(M + M^\dagger)\psi = 0$. Then*

$$M^\dagger = -M. \tag{12.3}$$

For real M, this is $M^T = -M$. The simplest 2×2 antisymmetric matrix J satisfies $J^2 = -I$ (so J acts as the imaginary unit), and

$$M = J \otimes L_Q \quad \Longrightarrow \quad i\partial_t \psi = L_Q\, \psi, \tag{12.4}$$

Intuition. *Norm conservation forces the generator to be anti-Hermitian; the block J with $J^2 = -I$ supplies the imaginary unit.* ***Meaning.*** *It is the unitary dynamics of the theory: the Schrödinger equation.* ***Why it matters.*** *It means quantum mechanics is* produced *from a conservation requirement, not postulated.* ***Limitations.*** *Linear and single-particle; the imaginary structure is imported through J.*

with unitary evolution

$$\psi(t) = e^{-iL_Q t}\, \psi(0). \tag{12.5}$$

Intuition. Conservation of the norm forces the generator to be anti-Hermitian; the simplest anti-Hermitian object on the graph is $J \otimes L_Q$, and $J^2 = -I$ is the imaginary unit in disguise. The "i" of quantum mechanics is thus not postulated but built from a 2×2 antisymmetric matrix.

Significance. This is the dynamical core of the theory: the Schrödinger equation is obtained from a conservation requirement plus the graph Laplacian, rather than asserted as an axiom.

Limitations. It is linear and single-particle. The imaginary structure is imported through the 2×2 block J, and the argument does not by itself reconstruct the field-theoretic, interacting content of quantum mechanics.

Historical context. The Schrödinger equation is normally *postulated*; the reduction program asked whether its "i" and its unitary evolution could instead be derived from a conservation requirement.

Why it mattered. If the dynamical core could be derived rather than assumed, the theory would *produce* quantum mechanics instead of importing it — a foundational, not cosmetic, gain.

What was tried before. The tight-binding/Hilbert-space route used L_Q as a generator, but the specific reduction of the imaginary unit to a 2×2 antisymmetric block J with $J^2 = -I$ was the step that had not been exploited.

Why the result was accepted. Accepted as a theorem (TQM-149–151) because it is exact, and because its continuum limit is then independently verified by `GraphLaplacianContinuumTests`.

12.4 Observables from the spectrum (TQM-145)

Observable	Definition
effective mass	$m_{\text{eff}} = 1/\lambda_1$
energy	$E = \text{tr}(L)$
gap	$\Delta = \lambda_2 - \lambda_1$
correlation length	$\xi = 1/\sqrt{\lambda_1}$
diffusion constant	$D = \lambda_1$
channel capacity	$C = \log_2 N$

Remark 12.4. *These are* dimensional assignments*, not dynamical predictions; the hostile reviewer correctly notes this is a scope boundary.*

12.5 The ontology → physics bridge

The bridge has two legs, both now tested:

1. L_Q eigenvector basis = Hilbert space (TQM-149); and
2. the causal-set continuum limit yields the metric (QG-001, XC006–012).

Part II — The Continuum-Limit Program (Verified)

This part records the two continuum chains of the theory and their (partial) connection. One chain is *elliptic*: the graph Laplacian L_Q converges to the flat Laplacian $-\nabla^2$, and its weighted generalization supplies a curved-space Schrödinger equation and a Laplace–Beltrami approximation. The other is *Lorentzian*: the causal-set BDG operator converges to the d'Alembertian $\Box$. The two chains are disjoint in signature, and the part's central fact is that no native bridge between them has yet been closed. The Schrödinger side is controlled and tested; the Einstein side is partial.

Key results. $L_Q = D - A$ is the generator and its eigenbasis is the Hilbert space; the Schrödinger equation $i\partial_t\psi = L_Q\psi$ follows from norm conservation.
Open questions. The spectrum-derived observables are dimensional assignments, not dynamical predictions.
Verification status. Mathematical (TQM-142 tight-binding identity; TQM-149–151 reversibility theorem); the continuum-limit tests in the next part verify the downstream chain.

Chapter 13

The Two Continuum Chains

The two chains share a substrate (the Q-events) but use different operators on it. This chapter exhibits both limits and then states the signature incompatibility that separates them.

13.1 $L_Q \to -\nabla^2$: The Flat Laplacian

Theorem 13.1 (Discrete spectrum of the chain Laplacian)**.** *The graph Laplacian of a 1D chain of N sites has the exact spectrum*

$$\lambda_k = \frac{1}{\Delta x^2}\left[2 - 2\cos\frac{\pi k}{N+1}\right], \qquad k = 1, \ldots, N, \tag{13.1}$$

with Δx the lattice spacing. In the continuum limit $N \to \infty$, $\Delta x \to 0$ with the length fixed,

$$\lambda_k \to (\pi k)^2. \tag{13.2}$$

Proof. The chain Laplacian is the tridiagonal Toeplitz matrix with diagonal $2/\Delta x^2$ and off-diagonal $-1/\Delta x^2$; its eigenvectors are the discrete sine modes $\sin(\pi k j/(N+1))$. □

Verification. `GraphLaplacianContinuumTests` builds chains with $N = 32, 64, 128, 256$, diagonalizes L_Q via MathNet.Numerics, and compares against the closed form. The relative error *decreases* with N (second-order convergence, error $\sim 4\times$ per doubling), establishing that L_Q converges to the flat Laplacian with controlled error. **Result: PASSED.**

Intuition. Standing waves on a chain are discrete sine modes; taking $N \to \infty$ at fixed length makes their dispersion relation the usual momentum-squared spectrum of a free particle on a line.

Significance. This is the quantitative bridge from the discrete ontology to the continuum: it is an exact closed form, not a numerical fit, and its convergence is controlled to second order.

Limitations. It is one-dimensional, unweighted, and elliptic only. It says nothing about the Lorentzian chain or about curvature.

Historical context. The question was whether the discrete ontology has a controlled continuum limit at all — the reviewer's "no controlled continuum limit" objection.

Why it mattered. A controlled limit is what separates a toy discrete model from a candidate for continuum physics; without it the Schrödinger program has no bridge to space.

What was tried before. Naive finite-difference limits were known, but a closed-form spectrum with a demonstrated second-order convergence ($\sim 4\times$/doubling) had not been established for L_Q.

Why the result was accepted. Accepted because it is an exact closed form (not a fit) and the error demonstrably decreases with N in `GraphLaplacianContinuumTests`.

13.2 BDG $\to \Box$: The d'Alembertian

The causal-set d'Alembertian is the Benincasa–Dowker–Glaser (BDG) operator: a directed, alternating sum over causal intervals of the sprinkling.

Theorem 13.2 (BDG convergence)**.** *The BDG operator converges to the continuum d'Alembertian* $\Box$ *as the sprinkling density grows, with leading truncation error* $O(h^2)$*.*

Verification. `BDGOperatorContinuumTests` verifies convergence with a tolerance 10^{-2} (the leading $O(h^2)$ error at $h = 1/16$ is $\approx 4.37 \times 10^{-3}$); the error falls $\sim 4\times$ per h-halving. **Result: PASSED.**

Intuition. A discretized wave operator on a causal set must sum alternatingly over past and future intervals to cancel the leading terms; the BDG prescription does exactly this, leaving the continuum d'Alembertian in the dense limit.

Significance. It establishes the Lorentzian half of the continuum program with controlled $O(h^2)$ error, matching the elliptic half's level of control.

Limitations. The operator is imported from causal-set theory rather than derived from the four primitives, and the tolerance is leading-order (10^{-2}), so only the lowest nontrivial terms are checked.

13.3 The Quantum-Gravity Bridge

The two continuum chains are *partially connected* — and their signatures are incompatible.

Proposition 13.3 (Two distinct operators)**.**

1. *L_Q is an* undirected, positive *graph Laplacian whose continuum limit is* $-\nabla^2$ *(Riemannian, elliptic, positive semi-definite).*
2. *The BDG operator is a* directed, alternating *causal-set operator whose continuum limit is* $\Box$ *(Lorentzian, hyperbolic, indefinite).*

Verification. `QuantumGravityBridgeTests` (three tests) establish:

1. L_Q is positive semi-definite (minimum eigenvalue ≥ 0);
2. the BDG operator is indefinite (possesses negative eigenvalues);
3. the two are incompatible in signature — the graph Laplacian cannot serve as a Lorentzian d'Alembertian (BdgUniquenessAnalyzer O3/O6 explicitly rejects it).

Result: PASSED (3/3).

Remark 13.4. *This is the single most important structural fact of the continuum program: TQM possesses* two *chain types, one Riemannian* ($L_Q \to -\nabla^2$) *and one Lorentzian* (BDG $\to \Box$)*, which are* disjoint *in signature. The bridge between them is not yet closed natively.*

Intuition. A positive semi-definite operator cannot be indefinite, so the elliptic and hyperbolic continuum limits cannot be the same object; the Q-events underdetermine which operator to use.

Significance. This fact is what makes the Einstein recovery *logical* rather than mathematical: the Schrödinger chain and the Einstein chain are separate, and their junction is the open problem.

Limitations. It states incompatibility, not a construction; it is a no-bridge result, not a bridge.

Key results. $L_Q \to -\nabla^2$ (exact, second-order) and BDG $\to \Box$ ($O(h^2)$) are both controlled; the two chains are disjoint in signature.
Open questions. No native bridge connects the elliptic and Lorentzian chains.
Verification status. `GraphLaplacianContinuumTests`, `BDGOperatorContinuumTests`, `QuantumGravityBridgeTests` (3) — all PASS.

Chapter 14

The Weighted Laplacian, Laplace–Beltrami, and Curved Schrödinger

Having established the flat and Lorentzian chains, this chapter turns to curvature. The weighted Laplacian L_W is the natural curved generalization of L_Q: it supplies a well-defined curved Schrödinger equation and a Laplace–Beltrami approximation, even though the metric that would make it a genuine Δ_g remains external.

14.1 The weighted Laplacian

Definition 14.1 (Weighted Laplacian). *Given the spatial coupling matrix K_{ij}, the weighted graph Laplacian is*

$$L_W = D_K - K, \qquad (D_K)_{ii} = \sum_j K_{ij}. \tag{14.1}$$

This is the standard unnormalized *graph Laplacian (Belkin–Niyogi), valid for uniform density.*

Proposition 14.2 (Properties). *L_W is symmetric, has zero row sum, is positive semi-definite, and reduces to the unweighted Laplacian when $K = A$.*

Verification. `WeightedLaplacianTests` (four tests) confirm symmetry, zero row sum, positive semi-definiteness, and reduction to the unweighted case. **Result: PASSED (4/4).**

Intuition. Weighting the edges of the graph lets the geometry be encoded in the couplings; the unnormalized Laplacian is the operator whose zero-energy state is uniform on the weighted graph.

Significance. L_W is the tool that carries the theory from the flat chain to curved space, and its basic properties are verified rather than assumed.

Limitations. The unnormalized Laplacian is the right operator only at uniform density; at non-uniform density the normalized variants differ, and none of this supplies the metric itself.

14.2 The Laplace–Beltrami approximation

Theorem 14.3 (Laplace–Beltrami from L_W). *L_W approximates the Laplace–Beltrami operator Δ_g in the flat limit and reproduces standard manifold examples.*

Verification. `LaplaceBeltramiTests` (three tests):

1. *Flat limit*: L_W preserves the flat spectrum;
2. *Variable weights*: non-uniform K changes the spectrum;

3. *Known manifold*: the S^1 cycle graph reproduces the circle spectrum.

A subtle degeneracy was found and fixed by audit: the nonzero S^1 modes are two-fold degenerate ($e^{\pm ik\theta}$), so in ascending order the first occurrence of mode k is at index $2k-1$, not k. **Result: PASSED (3/3).**

Intuition. The Laplace–Beltrami operator is "the Laplacian with the metric in it"; approximating it from a weighted graph means recovering the manifold's spectrum from the couplings.

Significance. It connects the graph operator to the standard geometric operator of curved space and exposes (and fixes) the S^1 degeneracy subtlety.

Limitations. It is an approximation, not a theorem for arbitrary g, and the metric that makes L_W a genuine Δ_g is still imported rather than generated.

14.3 The curved-space Schrödinger equation

Theorem 14.4 (Curved Schrödinger equation)**.** *The operator L_W defines a curved-space Schrödinger equation*

$$i\,\partial_t \psi = L_W \psi. \tag{14.2}$$

Because L_W is self-adjoint, the evolution is unitary and the norm is conserved:

$$\partial_t \|\psi\|^2 = 0. \tag{14.3}$$

Verification. `CurvedSchrodingerTests` (three tests): unitarity (norm conservation), reduction to the flat Schrödinger equation, and the existence of a curved operator. **Result: PASSED (3/3).**

Remark 14.5. *This establishes the* Schrödinger *side of the curved-space bridge: the weighted Laplacian L_W produces a well-defined, unitary curved Schrödinger equation. The* metric *that would make L_W a genuine Δ_g remains external (Chapter 16).*

Intuition. Self-adjointness is exactly what guarantees unitarity, so the curved equation inherits norm conservation for free.

Significance. It closes the Schrödinger side of the curved-space program: a well-defined, tested, unitary curved dynamics exists.

Limitations. The curved equation is formal until a metric is supplied; the metric side is external, which is the reason the Einstein side remains partial.

Key results. L_W is symmetric/PSD/zero-sum; it gives a Laplace–Beltrami approximation and a unitary curved Schrödinger equation.
Open questions. The metric that would make L_W a genuine Δ_g remains external.
Verification status. `WeightedLaplacianTests` (4), `LaplaceBeltramiTests` (3), `CurvedSchrodingerTests` (3) — all PASS.

Chapter 15

The Einstein Tensor Chain

This chapter exhibits the standard differential-geometry chain from a metric to the Einstein tensor. The chain is fully functional and tested; the point of including it is to show *exactly where* it breaks when one tries to feed it TQM's own objects.

15.1 Standard differential geometry

Definition 15.1 (Christoffel symbols)**.**

$$\Gamma^{\lambda}_{\mu\nu} = \tfrac{1}{2}\, g^{\lambda\sigma}\left(\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu}\right). \tag{15.1}$$

Definition 15.2 (Riemann curvature)**.**

$$R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu \Gamma^{\rho}_{\nu\sigma} - \partial_\nu \Gamma^{\rho}_{\mu\sigma} + \Gamma^{\rho}_{\mu\lambda}\Gamma^{\lambda}_{\nu\sigma} - \Gamma^{\rho}_{\nu\lambda}\Gamma^{\lambda}_{\mu\sigma}. \tag{15.2}$$

Definition 15.3 (Ricci and Einstein tensors)**.**

$$R_{\sigma\nu} = R^{\rho}{}_{\sigma\rho\nu}, \qquad G_{\mu\nu} = R_{\mu\nu} - \tfrac{1}{2} R\, g_{\mu\nu}. \tag{15.3}$$

15.2 Verified examples

Unit 2-sphere $g = \mathrm{diag}(1, \sin^2\theta)$ at $\theta = \pi/4$:

$$\Gamma^{\theta}_{\phi\phi} = -\sin\theta\cos\theta = -0.5, \qquad K = 1, \qquad R = 2K = 2, \qquad R_{\mu\nu} = K g_{\mu\nu}. \tag{15.4}$$

Unit 3-sphere $g = \mathrm{diag}(1, \sin^2\chi, \sin^2\chi\sin^2\theta)$ at $\chi = \theta = \pi/4$: $R_{\mu\nu} = 2g_{\mu\nu}$, $R = 6$, hence

$$G_{\mu\nu} = R_{\mu\nu} - \tfrac{1}{2}Rg_{\mu\nu} = -g_{\mu\nu} = -\mathrm{diag}(1, \tfrac{1}{2}, \tfrac{1}{4}). \tag{15.5}$$

Theorem 15.4 (2D Einstein tensor vanishes)**.** *In two dimensions $G_{\mu\nu} \equiv 0$ identically ($R_{\mu\nu} = \frac{1}{2}Rg_{\mu\nu}$); a non-trivial $G_{\mu\nu}$ requires dimension ≥ 3.*

Verification. `EinsteinTensorTests` (four tests: metric→Christoffel, Christoffel→Riemann, Riemann→Ricci, Ricci→Einstein) and `EinsteinTensorIntegrationTests` (four tests on the integrated builder `EinsteinTensorBuilder`). **Result: PASSED (8/8).**

Intuition. In two dimensions the Ricci tensor is forced to be proportional to the metric, so the traceless Einstein tensor vanishes identically; curvature dynamics need at least three dimensions to have room.

Significance. It explains, within standard geometry, why a dynamical Einstein equation lives in $d \geq 3$ — a fact the dimensionality program (Chapter 17) turns into a selection argument.

Limitations. It is standard differential geometry, not a TQM derivation; it does not by itself produce 3+1.

Remark 15.5 (Where the chain breaks in TQM)**.** *The standard chain is fully functional. But TQM's own analyzers (`QuantumGravityEmergenceAnalyzer`, `GrBridgeAnalyzer`)* describe *the chain as strings (`GeoStep`) and* never compute *it. The chain breaks at Step 1 —* metric → Christoffels *— because TQM possesses no native metric field $g_{\mu\nu}(x)$: it is imported (Chapter 16).*

Chapter 16

Metric Emergence and Origin Closure

This chapter asks how a metric can arise at all, and records the four-link chain that closes the metric *origin* — while leaving the metric *dynamics* imported.

16.1 The causal distance structure

Definition 16.1 (Causal volume and distance)**.** *For a causal interval of depth D in d dimensions, the causal volume is*

$$N(D) \propto D^d, \tag{16.1}$$

and the causal distance is

$$L = N^{1/d}. \tag{16.2}$$

Verification. `MetricGenerationTests` and `MetricEmergenceTests` confirm: $L = N^{1/d}$ recovers the depth exactly ($d = 2, 3, 4$; $D = 1, \ldots, 8$); the dimension is recovered from volume growth; and the resulting distance matrix satisfies all four metric axioms (non-negativity, identity of indiscernibles, symmetry, triangle inequality). **Result: PASSED.**

16.2 The conformal factor from the counting measure

Theorem 16.2 (Conformal factor)**.** *The counting measure fixes the volume element $\sqrt{|g|} = \rho$ (density ρ). For a conformally flat ansatz $g = f\,\eta$,*

$$\sqrt{|g|} = f^{\,d/2} = \rho \quad \Longrightarrow \quad f = \rho^{\,2/d}. \tag{16.3}$$

Intuition. *The event density is the natural volume element, and in a conformally flat ansatz the volume element fixes the conformal factor.* ***Meaning.*** *The counting measure determines the one remaining scalar degree of freedom.* ***Why it matters.*** *It closes the conformal* factor *natively, the native half of the metric-origin chain.* ***Limitations.*** *Conformally flat only; a general metric is not produced natively.*

Verification. `MetricEmergenceTests` confirms the round-trip $f^{d/2} = \rho$ for $\rho \in \{0.5, 1, 2, 8\}$, $d \in \{2, 3, 4\}$. The conformally flat candidate $g = f\eta$ is constructible: constant f gives $R = 0$; $f = 1 + \frac{1}{2}x^2$ gives $R = -0.6145$, matching the Liouville result $K = -\frac{1}{2f}\nabla^2 \ln f$. **Result: PASSED.**

Intuition. The density of events is the natural candidate for the volume element, and in a conformally flat ansatz the volume element alone fixes the conformal factor.

Significance. This closes the conformal *factor* natively: the counting measure determines the only remaining scalar degree of freedom.

Limitations. It assumes a conformally flat ansatz; a general metric is not produced natively, only its conformal class plus a flat representative.

16.3 The conformal class: Malament's theorem

Theorem 16.3 (Malament 1977; Hawking–King–McCarthy 1976)**.** *A bijection preserving the causal order* $\prec$ *between two past-and-future-distinguishing spacetimes is a conformal isometry: the causal order determines the metric* up to a conformal factor.

Verification. `ConformalStructureTests` and `MetricOriginTests` establish: the causal order is a valid partial order whose boundary is the light cone; the null structure is invariant under $g \to fg$ $(f > 0)$ but *changed* by a non-conformal rescale $g = \mathrm{diag}(-1, 2)$; hence the light cone picks out *exactly* the conformal class. **Result: PASSED.**

Intuition. Two observers who agree on "what is before what" agree on the light cone, hence on the metric up to an overall local rescale — the conformal class.

Significance. This is the imported theorem that turns the causal order into geometry; importing a proven theorem is standard practice, not a defect.

Limitations. It is imported, not TQM-derived, and it needs past-and-future-distinguishing spacetimes; it fixes only the conformal class, leaving the factor to be supplied separately.

Theorem 16.4 (Metric origin closure)**.** *The conformal-class "gap" is an* imported theorem *(Malament), not a theory gap. The metric origin is closed:*

$$\textit{Q-events} \to \textit{causal order (native)} \to \textit{conformal class (proven)} \to \textit{conformal factor (native)} \to g_{\mu\nu}. \tag{16.4}$$

Classification of the metric origin:

Link	Status	Nature
Q-events	**NATIVE**	TQM-derived primitive
Causal order	**NATIVE**	precedence from individuation
Conformal class	**IMPORTED**	Malament/HKM — *proven*
Conformal factor	**NATIVE**	counting measure, $f = \rho^{2/d}$
Full $g_{\mu\nu}$	**determined**	class × factor

Remark 16.5. *The metric origin is* already solved*: importing a proven theorem is standard, not a defect. The genuinely open item is the* native *metric* $\to$ *operator coupling (the "G4" gap): TQM does not produce* $g_{\mu\nu}$ *from Q-events, it imports it.*

Intuition. The metric is assembled from three native links and one proven import; the "gap" is a proven theorem, so the origin is genuinely closed.

Significance. It is the theory's answer to "where does geometry come from": a chain, not a postulate.

Limitations. Closing the origin does not close the dynamics: the native metric-to-operator coupling (G4) remains open, which is why Einstein recovery is logical rather than mathematical.

Historical context. The question was whether a metric $g_{\mu\nu}$ can arise from Q-events at all, and where the theory must stop.

Why it mattered. Geometry is the substrate of the Einstein side; a native origin would close the largest remaining gap in the continuum program.

What was tried before. Direct construction from Q-events failed (the metric-dependent operator is absent); the route that worked imports Malament's theorem for the conformal class and supplies the conformal factor natively.

Why the result was accepted. Accepted as closure of the metric *origin* — the import is a proven theorem, not a gap — while the metric *dynamics* (G4) remain honestly imported.

Key results. The metric origin is closed: Q-events $\to$ causal order $\to$ conformal class (Malament, proven) $\to$ conformal factor ($f = \rho^{2/d}$, native) $\to$ metric.
Open questions. The native metric-to-operator coupling (G4) remains open; metric dynamics are imported.
Verification status. `MetricGenerationTests` (4), `MetricEmergenceTests` (4), `ConformalStructureTests` (3), `MetricOriginTests` (3) — all PASS.

Part III — The Derivation Hierarchy

This part runs the derivation hierarchy from the continuum objects down to concrete physics: dimensionality, gauge structure, flavor, gravity, and cosmology. Each chapter records what is **DERIVED**, what is **REAL-UNDERIVED**, and what is **DRAWN**, with the confidence assessments that the hostile-review history refined.

Chapter 17

Complexity and Spatial Dimensionality

17.1 The complexity functional

The complexity argument is a **weighted six-component decomposition** of the conditions for observers (`ComplexityOptimumAnalyzer.cs`), not a single scalar variational functional.

Component	Weight	M^2 window
Structure formation	3.0	2–8
Particle stability	2.5	3–7
Chemistry potential	4.0	3–5
Information capacity	2.0	3–10
Evolution potential	1.5	2–6
Observer viability	5.0	4–6

17.2 Spatial dimensionality $d = 3 + 1$

Theorem 17.1 (Unique dimensionality). $d = 3 + 1$ *is the* unique *value satisfying simultaneously: Bertrand's theorem (closed orbits only for* $1/r^2$*), Gauss's law (*$F \propto 1/r^{d-1}$*, Coulomb only in 3D), knot stability (codimension-2), Huygens (sharp propagation in odd* d*), and 2 GR polarizations.* $d = 2{+}1$ *fails gravity/atoms/knots;* $d \geq 4{+}1$ *fails orbits and knots.*

Remark 17.2. $M^2 \approx 5$ *is the observer-viability* intersection *of the component windows, not a hand-inserted number. Confidence T-02 = 0.85; this is a window-intersection argument, not a variational theorem. The analyzer records that the generation count* $G \approx 3$ *is a* plateau*, not a peak.*

Intuition. Each requirement — closed orbits, Coulomb forces, knots, sharp signals, gravitational polarizations — is a filter on dimension, and their intersection is a single value.

Significance. Dimensionality is derived rather than assumed, closing one of the largest "why" questions with a testable-style intersection argument.

Limitations. It is a window-intersection argument (confidence 0.85), not a variational theorem, and it derives only the *spacetime* 3; the *internal* 3 (generations, color) is a separate, selected quantity (Chapter 23).

Key results. $d = 3{+}1$ is the unique value satisfying Bertrand, Gauss, knot stability, Huygens, and the two GR polarizations simultaneously; $M^2 \approx 5$ is the observer-viability window.
Open questions. The argument is a window-intersection, not a variational theorem (T-02 = 0.85).
Verification status. Analysis via `ComplexityOptimumAnalyzer`; not part of the executable test inventory.

Chapter 18

Gauge Structure

This chapter records the gauge program's outcome: $U(1)$ derived, $SU(2)$ emergent, $SU(3)$ contingent. The structure is derivable; the count is not.

18.1 $U(1)$ — derived

Theorem 18.1 ($U(1)$ from the circle)**.** *Phase lives on S^1. Its isometry group is*

$$\mathrm{Aut}(S^1) = U(1), \qquad \pi_1(S^1) = \mathbb{Z}, \tag{18.1}$$

Intuition. *Phase is an angle, so it lives on a circle; the circle's symmetries are $U(1)$, and going around it an integer number of times is a conserved charge.* ***Meaning.*** *$U(1)$ is the symmetry of phase itself.* ***Why it matters.*** *It removes a free gauge assumption from the standard toolkit.* ***Limitations.*** *Structure only: the Maxwell action is still borrowed, not produced.*
and the winding $\oint \nabla\theta \cdot dl = 2\pi n$ yields integer topological charge. (Confidence 0.95.)

Intuition. Phase is an angle, so it lives on a circle; the symmetries of a circle are $U(1)$, and going around the circle an integer number of times is a conserved charge.

Significance. This is the cleanest result of the gauge program: $U(1)$ is not chosen but is the symmetry of phase itself, promoted to a theorem (confidence 0.95).

Limitations. It derives the group structure, not the gauge dynamics: the Maxwell action is borrowed, not produced.

Historical context. The question was whether $U(1)$ electromagnetism is a separate input or follows from the ontology of phase.

Why it mattered. $U(1)$ is the oldest gauge symmetry; deriving it from the circle would remove a free assumption from the standard toolkit.

What was tried before. Group-theory approaches *postulate* $U(1)$; the defect/winding route tried to *derive* it from $\mathrm{Aut}(S^1)$ and $\pi_1(S^1) = \mathbb{Z}$.

Why the result was accepted. Accepted as the cleanest result of the gauge program (confidence 0.95): not a choice but the symmetry of phase itself — while admitting the Maxwell action is still borrowed.

18.2 $SU(2)$ and $SU(3)$ — real-underived structure

Factor	Status	Confidence
$U(1)$	**DERIVED**	0.95
$SU(2)$	**REAL-UNDERIVED** (emergent)	0.70
$SU(3)$ structure	**REAL-UNDERIVED** (emergent)	0.10
color count $n = 3$	**DRAWN**	—

Binary winding $\{n = \pm 1\}$ gives the minimal doublet; the Bloch sphere gives $SO(3) = SU(2)/Z_2$; the spinor double cover gives $SU(2)$. The automorphism of the n-defect moduli space satisfies $\mathrm{Aut}(C^n/S_n) \supseteq SU(n)$, deriving $SU(3)$ *structure* but not the count.

Remark 18.2. *TQM derives the gauge-group* ***structure*** *(which group),* ***not*** *the gauge* ***dynamics*** *(Maxwell / Yang–Mills actions are borrowed; the 8-gluon algebra is assumption A-07). The count $n = 3$ is unfixed (gauge-count no-go T-09, provisional 0.10).*

Key results. $U(1)$ is derived ($\mathrm{Aut}(S^1) = U(1)$, $\pi_1(S^1) = \mathbb{Z}$); $SU(2)$ is emergent; $SU(3)$ is contingent (structure, not dynamics).
Open questions. The gauge count $n = 3$ is underived (provisional no-go T-09, 0.10).
Verification status. Mathematical (topology); confidence 0.95 / 0.70 / 0.10; no dedicated executable test.

Chapter 19

Flavor and the Koide Relation

The flavor chapter is the clearest single illustration of the structure/content split: the operator *form* is derived, its *spectrum* is contingent, and the Koide relation sits exactly on the boundary — real, precise, predictive, and underivable.

19.1 The Yukawa operator

The Yukawa operator is the overlap operator

$$Y_{ij} = \langle \mathrm{arch}_i \,|\, \mathrm{amplitude} \,|\, \mathrm{arch}_j \rangle. \tag{19.1}$$

Its *form* is derived; its *spectrum* (the hierarchy 1:207:3478) is underived — set by the contingent architecture shapes.

19.2 The Koide relation

Theorem 19.1 (Koide relation)**.** *The charged-lepton pole masses satisfy*

$$Q = \frac{\sum m}{(\sum \sqrt{m})^2} = 0.6666605, \qquad \theta = 44.9997^\circ, \tag{19.2}$$

Intuition. *In $\sqrt{m}$ (amplitude) space the three charged leptons sit at 45° to the democratic direction.* ***Meaning.*** *$Q = 2/3$ is $1/(3\cos^2 45^\circ)$; the relation is about amplitudes, not masses.* ***Why it matters.*** *It is the sharpest numerical clue in the flavor sector: real, predictive, RG-stable.* ***Limitations.*** *Its origin is underivable (no-go T-08): real does not mean derived.*

equivalently $Q = 2/3$ at $\theta = \arccos(1/\sqrt{2}) = 45^\circ$. The relation is real ($\sim 10^{-5}$ precision; Bayes factor vs. coincidence $\approx 3.2 \times 10^4$), predictive (1981 prediction $m_\tau = 1776.97$ MeV, confirmed 1992), and RG-stable.

Intuition. In $\sqrt{m}$ (i.e., amplitude) space the three charged leptons sit at a 45° angle to the democratic direction, which is exactly the condition $Q = 2/3$; the relation is about amplitudes, not masses.

Significance. It is a real, predictive, RG-stable regularity — the strongest numerical clue in the flavor sector — and the canonical example of a **REAL-UNDERIVED**structure.

Limitations. Its origin is underivable (no-go T-08 below): real does not mean derived, and the relation does not by itself fix the third-generation count.

Historical context. The Koide relation has stood since 1981 as an unexplained, high-precision regularity; the program asked whether its origin is derivable from the primitives.

Why it mattered. It is the single sharpest numerical clue in the flavor sector; its origin would either confirm a deep structure or prove that structure stops at form.

What was tried before. Seven routes were tried — symmetry, topology, attractors, information geometry, group theory, $m = 3$ closure, moduli — and all failed or blocked.

Why the result was accepted. Accepted as a *closed question* (no-go T-08): real, predictive, RG-stable, and underivable — the canonical **REAL-UNDERIVED**structure, with the neutrino-Koide analogue falsified (T-11).

No-Go Theorem 19.2 (Koide origin — T-08)**.** *No symmetry (S_3), attractor, topology (S^1), or information-geometry selects $Q = 2/3$. (Confidence 0.70.)*

The closure audit (Phase 159) exhausted seven routes:

Route	Status
Symmetry (S_3)	No-Go
Topology (S^1)	No-Go
Attractors	No-Go
Information Geometry	No-Go
Group Theory	No-Go
$m = 3$ Closure	No-Go
Moduli Arguments	Blocked

The Koide origin is a *closed question* — real, predictive, RG-stable, *underivable* — the canonical **REAL-UNDERIVED** structure.

No-Go Theorem 19.3 (Neutrino-Koide — T-11, falsification)**.** *Requiring $Q = 2/3$ for neutrinos gives $Q_{\max} =$ 0.585 (normal) / 0.500 (inverted), both $< 2/3$; the measured Δm^2 exclude all-lepton Koide. (Confidence 0.90.)*

Key results. The Koide relation is real, predictive, and RG-stable; its origin is underivable (T-08); the neutrino-Koide analogue is falsified (T-11).
Open questions. None: the Koide origin is a closed question, and the falsification closed the last distinguisher.
Verification status. Computational audits (Q, $Q_{\max}$); recorded in the Koide closure, not in the executable test inventory.

Chapter 20

Gravity and Emergent GR

Gravity in TQM is a phase-gradient phenomenon. This chapter records the honest status: the Einstein recovery is *logical*, not a new dynamical derivation.

20.1 Newton's constant

Proposition 20.1 (Dimensional result). *$G = \ell^2 c^3/\hbar$ is a* unit-consistency *result, not a derivation of gravity's dynamics.*

Intuition. The triple $(\ell, \tau, \hbar)$ can assemble exactly one combination with the dimensions of G. **Meaning.** G is fixed by units once the scale ℓ is fixed. **Why it matters.** It removes Newton's constant from the list of independent inputs. **Limitations.** Dimensional analysis only; it says nothing about the field equations.

Intuition. The triple $(\ell, \tau, \hbar)$ can assemble exactly one combination with the dimensions of G, so Newton's constant is fixed by units once the scale is fixed.

Significance. It removes G from the list of independent constants — gravity's strength is a bookkeeping consequence of the scale triple.

Limitations. It is dimensional analysis, not dynamics; it says nothing about the field equations.

20.2 The phase-gravity chain

oscillation → phase → causal density → metric → curvature → Einstein.

Theorem 20.2 (Leading-order recovery).

$$G_{\mu\nu} = 8\pi G_{\text{eff}}\, T_{\mu\nu} + O(\ell_P^2 R^2), \tag{20.1}$$

with Planck-scale corrections $\sim 10^{-40}$. Newtonian $1/r^2$, lensing, redshift, perihelion precession, and two gravitational-wave polarizations at speed c are reproduced.

Two modifications beyond GR: (1) time-varying dark energy

$$\Lambda(t) = \frac{\alpha}{\sqrt{V(t)}}, \tag{20.2}$$

where $V(t)$ is the 4-volume of the past light cone; and (2) singularity resolution at $r \sim \ell_P$ (maximum curvature $1/\ell_P^2$).

Remark 20.3 (Honesty note). *The chain is **logical, not a new dynamical derivation**: "same equations, same predictions; no falsifiable difference" in the weak field. The value is* ontological *(identifying the gravitational potential with the phase field); the physical content beyond GR is confined to $\Lambda(t)$ and singularity resolution.*

Intuition. If the phase field is the gravitational potential, then the standard curvature chain is recovered to leading order, with only Planck-scale corrections — gravity is re-labeled, not replaced.

Significance. It shows the ontological fit is consistent with GR in the weak field while isolating the two places where TQM genuinely departs ($\Lambda(t)$ and singularity resolution).

Limitations. It is logical, not mathematical: the metric and BDG action are imported, so this is a reinterpretation, not a derivation of Einstein's equations.

20.3 The radial acceleration relation

Theorem 20.4 (Zero-parameter RAR)**.**

$$g_\dagger = \frac{cH_0}{2\pi} \approx 1.05 \times 10^{-10}\ \mathrm{m/s}^2, \tag{20.3}$$

Intuition. *cH_0 is the natural acceleration scale of a causal set; dividing by 2π reproduces the Milgrom scale a_0.* ***Meaning.*** *It is a zero-parameter prediction that lands on the measured value.* ***Why it matters.*** *It is the theory's sharpest dark-sector result.* ***Limitations.*** *The 2π is an admitted numerical accident; this is a match, not yet a full derivation.*

matching the measured a_0. (The factor 2π is a numerical accident, Phase 144.)

Intuition. The combination cH_0 is the natural acceleration scale of a causal set; dividing by 2π reproduces the observed Milgrom scale a_0.

Significance. It is a zero-parameter prediction that lands on the measured value, a genuine sharp result in the dark-sector.

Limitations. The 2π factor is a numerical accident, and the relation is a match, not yet a full derivation of the dynamics behind a_0.

Key results. $G = \ell^2 c^3/\hbar$ is dimensional; the leading-order Einstein recovery is logical; the RAR $g_\dagger = cH_0/2\pi$ matches a_0 with zero free parameters.
Open questions. Native metric dynamics (G4); a unique, sharp discriminating prediction.
Verification status. `EinsteinTensorTests` (4) and `EinsteinTensorIntegrationTests` (4) verify the standard chain; `EinsteinRecoveryTests` (3) = PARTIAL.

Chapter 21

Cosmology

Cosmology is the least closed chapter of the monograph: it is an accepted partial computational layer, not a completed derivation. The status is recorded without overclaim.

- **Expansion** is an interpretation of FLRW (QG-081); every reinterpretation collapses to FLRW (QG-080–089).
- **Causal-set** Λ: $\Lambda \sim 1/\sqrt{N}$ is a postdiction (Phase 140).
- **Pantheon+SH0ES**: TQM and ΛCDM are indistinguishable at the current signal $w(z) = -1+0.015(1+z)^{3/2}$ (DATA-001); a detectability audit (DATA-002) quantified the power required for separation.
- **Clusters**: Coma dynamical mass and ACCEPT X-ray gas fraction are reproduced.
- **CMB**: an accepted partial computational layer; full C_ℓ solver deferred (computational, not a theory gap).

Remark 21.1. *The cosmology status is deliberately partial: the background and the compression peaks of the CMB are present, but the rarefaction peak and the acoustic phase shift require a CAMB-class Boltzmann solver — standard ΛCDM physics, not new TQM physics.*

Key results. Expansion is an FLRW reinterpretation; causal-set $\Lambda \sim 1/\sqrt{N}$ is a postdiction; the $w(z)$ signal is quantified.
Open questions. The full CMB C_ℓ solver is deferred (computational, not a theory gap).
Verification status. Partial computational layer (DATA-001/002, CMB closure audits); not in the executable test inventory.

Part IV — Classification, No-Gos, and Predictions

Chapter 22

The Three-Category Taxonomy

The taxonomy is the theory's honest-accounting device. It classifies *underivability*, so that no claim is allowed to pass as a derivation when it is a classification.

Definition 22.1 (Taxonomy)**.**

Category	*Meaning*
DERIVED	*computable from the primitives by theorem*
REAL-UNDERIVED	*real, precise, predictive structure whose origin is not computable*
DRAWN	*a coincidental draw of the abundance law*

The taxonomy classifies underivability*;* ***REAL-UNDERIVED*** *and* ***DRAWN*** *are* not *claimed as derivations.*

22.1 The fourteen classified results

Object	Category	Confidence
$U(1)$	**DERIVED**	0.95
spatial 3	**DERIVED**	0.85
$N \geq 3$	**DERIVED**	0.90
log-normal law	**DERIVED**	theorem
$SU(2)$	**REAL-UNDERIVED** (emergent)	0.70
$SU(3)$ structure	**REAL-UNDERIVED** (emergent)	0.10
Koide $Q = 2/3$ (reality)	**REAL-UNDERIVED** (structured)	0.90
Koide 45° (origin)	**REAL-UNDERIVED** (structured)	0.70
Yukawas / couplings / Ω_{DM}	**DRAWN**	—
$N \leq 3$	**DRAWN**	0.70
color count 3	**DRAWN**	—
internal $N = 3$	**DERIVED**∩**DRAWN**	0.70
neutrino-Koide	FALSIFIED	0.90

Category conflicts: **0**. Composite objects: **2**. Overall consistency: **0.95**.

Chapter 23

The No-Go Theorems

The no-go theorems are *conditional* — relative to the no-new-primitives constraint and the tested route space — not absolute logical impossibilities (except the computation T-11).

Theorem	Statement	Confidence
T-08 Koide no-go	no symmetry/attractor/topology/information-geometry selects $Q = 2/3$	0.70
T-09 Gauge-count no-go (provisional)	no principle fixes n; graph-spectrum/lattice-mode untested	0.10
T-10 $N \leq 3$ no-go	no stability/anomaly/representation bound gives $N \leq 3$	0.70
T-11 Neutrino-Koide falsification	$Q_{\max} = 0.585/0.500 < 2/3$ (computation)	0.90
T-12 Shared-cascade no-go	one vs three cascades untestable from one universe	0.55

Remark 23.1 (Internal-3 disposition)**.** *Gauge count $n = 3$ and multiplicity $N \leq 3$ are one node with two formally unlinked faces (N vs n; assumption A-10). It is dispositioned* unresolved-contingent*: the multiplicity face closed (T-10, 0.70); the gauge-count face* provisionally *closed (T-09, 0.10). This is the theory's one open door.*

Intuition. Each no-go is a statement about what the tested route space cannot reach, not about what is logically impossible; the "3" that pervades physics reduces to one node with two faces.

Significance. The no-gos are the program's most distinctive product: they locate the exact boundary between the derivable and the contingent, which is the content of the structure/content split.

Limitations. They are conditional (relative to no-new-primitives and the tested routes), the gauge-count no-go T-09 is only provisional (0.10), and the shared-cascade no-go T-12 is weak (0.55).

Key results. T-08–T-12 are conditional no-gos; the internal-3 node is the theory's one open door.
Open questions. The gauge-count face (T-09, 0.10) is only provisionally closed.
Verification status. T-11 is a computation (falsification); T-08 is route exhaustion; the rest are recorded audits.

Chapter 24

Quantitative Predictions

The theory's claim to be scientific rests on the fact that it makes predictions that can fail, and has already failed one. The table records the live predictions and the one falsified prediction; the falsification is kept in the table rather than buried, because it is the strongest evidence that the program is not immunized.

Prediction	Type	Status
RAR $g_\dagger = cH_0/(2\pi) \approx 1.05 \times 10^{-10}$	zero-parameter	matches a_0
$w(z) = -1 + 0.015(1+z)^{3/2}$	cosmological	Euclid $> 3\sigma$ by $\sim$2030
$\Lambda(t) = \alpha/\sqrt{V(t)}$	dark energy	Euclid / Roman
log-normal abundance law	distribution form	testable
$N \geq 3$ (CP lower bound)	a priori	confirmed
neutrino-Koide $Q = 2/3$	falsifiable	**falsified**

The theory is *falsifiable*: it made a concrete prediction (neutrino-Koide) that was falsified, and it carries live quantitative predictions that distinguish it from SM +ΛCDM in the *form* sector, even though the *content* is DRAWN.

Chapter 25

Scope and Limitations

A theory that states its limitations is more useful than one that hides them. The following list is the theory's own accounting of where it stops; it is reproduced from the hostile-review response and is intended to be read as part of the theory, not as an apology for it.

1. The gauge-count face of the internal-3 node is only provisionally closed (T-09, 0.10).
2. Encyclopedia completeness ≈ 81%; the CMB chapter is a documented partial computational layer.
3. Content is contingent by design; the theory predicts the distribution *shape*, not the *values*.
4. The shared-cascade question is logically (not empirically) closed.
5. The unified action is a roadmap, not a result.
6. Confidence numbers are uncalibrated ordinal ranks (Phase-149/156 estimates).
7. The phase-gravity chain is ontological in the weak field.

Part V — Verification

Chapter 26

The Executable Verification Program

The claim that this is a *tested* theory rests on the executable program summarized here and inventoried in full in Appendix A. Every "PASS" below is the output of an xUnit test, not a hand computation.

26.1 Summary of verified results

Chain link	Test file	Result
$L_Q \to -\nabla^2$	GraphLaplacianContinuumTests	exact spectrum, 4×/doubling
BDG$\to \Box$	BDGOperatorContinuumTests	$O(h^2)$, 4×/halving
L_Q vs BDG signature	QuantumGravityBridgeTests (3)	PSD vs indefinite
no native Δ_g	CurvedSpaceBridgeTests (3)	bridge absent
weighted Laplacian	WeightedLaplacianTests (4)	symmetric/PSD/zero-sum
$L_W \to \Delta_g$	LaplaceBeltramiTests (3)	S^1 example
curved Schrödinger	CurvedSchrodingerTests (3)	unitary
Einstein recovery	EinsteinRecoveryTests (3)	PARTIAL
Einstein chain	EinsteinTensorTests (4)	standard 2D/3D
Einstein integration	EinsteinTensorIntegrationTests (4)	builder works
metric generation	MetricGenerationTests (4)	distance PRESENT
metric emergence	MetricEmergenceTests (4)	factor native
conformal structure	ConformalStructureTests (3)	class imported
metric origin	MetricOriginTests (3)	closure

26.2 The decisive findings

1. The *Schrödinger* side of the continuum limit is **controlled and tested**: $L_Q \to -\nabla^2 \to$ Schrödinger (and the curved L_W, and the Laplace–Beltrami approximation) are verified.
2. The *Einstein* side is **partially** verified: the standard $g \to G_{\mu\nu}$ chain works, but TQM has no native metric (Malament import), and the BDG action is imported.
3. The conformal-class "gap" is an *imported theorem* (Malament), not a theory gap.

Key results. The Schrödinger side of the continuum limit is controlled and tested; the Einstein side is partial; the conformal class is an imported proven theorem.
Open questions. The native metric-to-operator coupling (G4).
Verification status. 15 test files / 47 tests, all PASS (`EinsteinRecoveryTests` noted PARTIAL).

Chapter 27

The Hostile-Review Audit Trail

This chapter is the tabular record of the hostile-review history narrated in Chapter 5. It lists the Round-2 “FATAL” issues and their final classifications.

A hostile referee (Round 2) raised twelve FATAL issues. Each was re-evaluated against the executable tests. The classifications are:

Round-2 FATAL issue	Classification
Primitives undefined as mathematical objects	PARTIALLY RESOLVED
Schrödinger derivation circular/incomplete	RESOLVED
Gauge derivations = elementary topology	PARTIALLY RESOLVED
Complexity argument not a derivation	PARTIALLY RESOLVED
Structure/content split = immunization	PARTIALLY RESOLVED
Composite objects admitted	PARTIALLY RESOLVED
Internal-3 unresolved while “complete”	RESOLVED
T-09 at 0.10 cannot close	RESOLVED
“Structurally complete” overstatement	RESOLVED
Gravity→Einstein is ontological	RESOLVED
$\mathrm{Aut}(S^1) = U(1)$ as EM derivation	RESOLVED
No controlled continuum limit	PARTIALLY RESOLVED

Tally: RESOLVED = 6 · PARTIALLY RESOLVED = 6 · OPEN = 0.

27.1 What changed

The framing overstatements (“closed” vs. “dispositioned”, “no-go” vs. “conditional no-go”, “derivation” vs. “ontological reinterpretation”) were corrected in the revised paper. The *substantive* objection — “no controlled continuum limit” — was met on the Schrödinger side (now tested: $L_Q \to -\nabla^2 \to$ Schrödinger, and curved L_W, Laplace–Beltrami, unitary), and *partially* on the Einstein side (the standard chain works, but the metric and BDG action are imported).

27.2 The decisive residual

The single genuinely-open item is the **native metric → operator coupling** (the “G4” gap): TQM imports $g_{\mu\nu}$ (Malament) rather than generating it from Q-events. The metric *origin* is closed (Chapter 16); the metric *dynamics* remain imported.

Verdict: READY AS A FOUNDATION MONOGRAPH — READY AS A RESEARCH-PROGRAM PAPER — NOT READY AS A DERIVATION PAPER (see Chapter 29).

Chapter 28

What Survived Hostile Review

This chapter consolidates the two hostile-review rounds into a single survival table. For each major claim, it records the original status, the reviewer's criticism, and the final status after re-evaluation against the completed tests. The pattern is uniform: nothing was retracted, but many overstatements were downgraded to precise, defensible forms.

Claim	Original status	Review criticism	Final status
Primitives	"fully formalized"	undefined as mathematical objects	partially formalized (measure/action admitted missing)
Schrödinger equation	derived	circular / incomplete	derived and tested ($L_Q \to -\nabla^2 \to$ Schrödinger, unitary)
Gauge group	derived	elementary topology	$U(1)$ derived; $SU(2)$ emergent; $SU(3)$ contingent (structure, not dynamics)
Dimensionality 3+1	derivation	not a derivation	window-intersection (T-02 = 0.85)
Structure/content split	organizing principle	immunization	retained; falsifiability demonstrated (neutrino-Koide)
"Structurally complete"	complete	overstatement	"dispositioned"; internal-3 flagged open
No-go theorems	no-go	too strong	conditional no-go (relative to tested route space)
Gravity $\to$ Einstein	derivation	ontological, not derivation	ontological reinterpretation in the weak field
$G = \ell^2 c^3/\hbar$	derived	dimensional analysis	dimensional / unit-consistency result
Continuum limit	controlled	no controlled limit	Schrödinger side controlled and tested; Einstein side partial
Metric origin	closed	a gap	origin closed (Malament import, proven); dynamics imported (G4)
Unique prediction	testable	no discriminating prediction	live predictions; none yet unique vs. SM+ΛCDM
Koide relation	real structure	numerology	real, predictive, RG-stable; origin underivable (T-08)
Gauge count $n = 3$	derived	unfixed	provisional no-go T-09 (0.10)

Reading of the table. Every "derived" claim that survived hostile review became either a tested theorem ($U(1)$, the Schrödinger limit) or an explicitly scoped statement (window-intersection, ontological reinterpretation, unit-consistency). Every "gap" alleged by the reviewers resolved into one of two honest categories: an imported proven theorem (Malament) or an admitted theory limit (G4, the gauge count, the discriminating prediction). Nothing in this table was retracted, and nothing was silently dropped.

Key results. No claim was retracted; overstatements were downgraded to precise forms; the single genuine theory limit (G4) and the three derivation-paper blockers are now explicit.
Open questions. The same three scientific blockers (native metric coupling, unique prediction, native BDG action), which affect only the derivation-paper claim.
Verification status. Tabulated from the completed tests and the two hostile-review rounds; no executable test applies to the table itself.

Chapter 29

Conclusion

THE Q-MODEL is *structurally complete* in the precise sense that every in-scope question is **dispositioned** — derived, underivable, underdetermined, contingent, or accepted as a computational layer.

The single residual of genuine scientific tension is the **internal-3 node** — why the internal multiplicity/count saturates at 3 — dispositioned unresolved-contingent with a provisional gauge-count no-go (T-09, 0.10). This is the theory's one open door.

Structure is derivable. Content is realized.

Publication status: READY AS A FOUNDATION MONOGRAPH — READY AS A RESEARCH-PROGRAM PAPER — NOT READY AS A DERIVATION PAPER. The Schrödinger dynamical core is executable and tested; the Einstein recovery remains logical, not mathematical (imported metric + imported BDG action, G dimensional, no unique discriminating prediction). The foundation-monograph and research-program claims are unaffected by the remaining derivation-paper blockers, because a foundation monograph may legitimately use imported proven theorems.

Appendix A

Complete Test Inventory

Test file	Tests	Verified quantity	Status
GraphLaplacianContinuumTests	1	$\lambda_k = (1/\Delta x^2)[2 - 2\cos(\pi k/(N{+}1))] \to (\pi k)^2$	PASS
BDGOperatorContinuumTests	1	BDG$\to \Box$, $O(h^2)$	PASS
QuantumGravityBridgeTests	3	L_Q PSD; BDG indefinite; incompatible	PASS
CurvedSpaceBridgeTests	3	no Δ_g; $\Delta_g \to \nabla^2$; absent	PASS
MetricOperatorTests	4	candidate constructions	PASS
WeightedLaplacianTests	4	symmetric; zero-sum; PSD; unweighted limit	PASS
LaplaceBeltramiTests	3	flat limit; weights; S^1 ($2k{-}1$ degeneracy)	PASS
CurvedSchrodingerTests	3	unitary; flat limit; curved operator	PASS
EinsteinRecoveryTests	3	PARTIAL (no full $G_{\mu\nu}$)	PASS
EinsteinTensorTests	4	flat/sph Γ, K, R, G	PASS
EinsteinTensorIntegrationTests	4	builder; 3-sphere $G = -g$	PASS
MetricGenerationTests	4	distance PRESENT; candidate PARTIAL	PASS
MetricEmergenceTests	4	factor $f = \rho^{2/d}$; $R = -0.6145$	PASS
ConformalStructureTests	3	axioms; null invariant; Malament	PASS
MetricOriginTests	3	class unique; factor free; imported	PASS

Appendix B

Confidence and Classification Tables

The confidence numbers are uncalibrated ordinal ranks (Phase-149/156 estimates), not posterior probabilities. They rank, they do not measure.

B.1 Confidence table

Result	Confidence	Note
program consistency	0.95	zero category conflicts
$U(1)$	0.95	theorem
$N \geq 3$	0.90	CP lower bound
neutrino-Koide	0.90	falsification (computation)
spatial 3	0.85	window intersection
$SU(2)$	0.70	emergent
Koide 45°	0.70	structured
$N \leq 3$	0.70	no-go
$SU(3)$ structure	0.10	provisional no-go

B.2 Final classification

Question	Classification
Is the metric origin closed?	YES (imported Malament, proven)
Is the Schrödinger continuum limit controlled?	YES (tested)
Is the Einstein recovery a derivation?	NO (logical, imported)
Is TQM journal-ready?	NO
Is TQM white-paper-ready?	YES

Appendix C

Worked Derivation: The Einstein Chain on the 2-Sphere

This appendix exhibits the standard differential-geometry chain explicitly, so the reader can follow every step without opening the repository code.

C.1 Setup

The unit 2-sphere has metric (coordinates $x^1 = \theta$, $x^2 = \phi$)

$$g = \mathrm{diag}(g_{\theta\theta},\ g_{\phi\phi}) = \mathrm{diag}\big(1,\ \sin^2\theta\big), \qquad g^{\theta\theta} = 1, \quad g^{\phi\phi} = \frac{1}{\sin^2\theta}. \tag{C.1}$$

C.2 Christoffel symbols

$$\Gamma^{\lambda}_{\mu\nu} = \tfrac{1}{2} g^{\lambda\sigma}\left(\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu}\right). \tag{C.2}$$

The only non-vanishing partial derivative is $\partial_\theta g_{\phi\phi} = 2\sin\theta\cos\theta$. Therefore

$$\Gamma^{\theta}_{\phi\phi} = -\tfrac{1}{2}\, g^{\theta\theta}\, \partial_\theta g_{\phi\phi} = -\sin\theta\cos\theta, \tag{C.3}$$

$$\Gamma^{\phi}_{\theta\phi} = \Gamma^{\phi}_{\phi\theta} = \tfrac{1}{2}\, g^{\phi\phi}\, \partial_\theta g_{\phi\phi} = \frac{\cos\theta}{\sin\theta} = \cot\theta. \tag{C.4}$$

At $\theta = \pi/4$: $\Gamma^{\theta}_{\phi\phi} = -1/2$, $\Gamma^{\phi}_{\theta\phi} = 1$.

C.3 Riemann curvature

$$R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu \Gamma^{\rho}_{\nu\sigma} - \partial_\nu \Gamma^{\rho}_{\mu\sigma} + \Gamma^{\rho}_{\mu\lambda}\Gamma^{\lambda}_{\nu\sigma} - \Gamma^{\rho}_{\nu\lambda}\Gamma^{\lambda}_{\mu\sigma}. \tag{C.5}$$

The single independent component is

$$\begin{aligned} R^{\theta}{}_{\phi\theta\phi} &= \partial_\theta \Gamma^{\theta}_{\phi\phi} - \Gamma^{\theta}_{\phi\phi}\Gamma^{\phi}_{\theta\phi} && \text{(C.6)}\\ &= -\big(\cos^2\theta - \sin^2\theta\big) + \sin\theta\cos\theta \cdot \cot\theta && \text{(C.7)}\\ &= \sin^2\theta. && \text{(C.8)} \end{aligned}$$

At $\theta = \pi/4$: $R^{\theta}{}_{\phi\theta\phi} = 1/2$.

C.4 Gauss curvature, Ricci, and Einstein

Gauss curvature:

$$K = \frac{R^{\theta}{}_{\phi\theta\phi}}{g_{\phi\phi}} = \frac{\sin^2\theta}{\sin^2\theta} = 1. \tag{C.9}$$

Ricci tensor (in 2D, $R_{\mu\nu} = K g_{\mu\nu}$):

$$R_{\theta\theta} = K g_{\theta\theta} = 1, \qquad R_{\phi\phi} = K g_{\phi\phi} = \sin^2\theta = \tfrac{1}{2} \text{ at } \theta = \pi/4. \tag{C.10}$$

Ricci scalar:

$$R = g^{\mu\nu} R_{\mu\nu} = 2K = 2. \tag{C.11}$$

Einstein tensor (vanishes identically in 2D):

$$G_{\mu\nu} = R_{\mu\nu} - \tfrac{1}{2} R g_{\mu\nu} = 0. \tag{C.12}$$

All of these are reproduced numerically by `EinsteinTensorBuilder` to within 10^{-3}. A non-trivial $G_{\mu\nu}$ requires dimension ≥ 3; the unit 3-sphere gives $G_{\mu\nu} = -g_{\mu\nu}$ (Appendix A).

Appendix D

Research Programs

The derivation program is organized into research programs, each closed by executable experiments. The four programs that carry the technical weight of this monograph — the gauge, Koide, continuum, and metric programs — are narrated in Chapter 6; this appendix records the program ledger.

Program	Scope	Experiments
DATA	Cosmology & RAR	10 (DATA-001→010)
QM	Quantum foundations	5 (QM-001→005)
QG	Quantum gravity	31 (QG-001→031)
XC	Continuum-limit bridge	(audits + tests)

Key results of the full program:

- G is not fundamental: $G = \ell^2 c^3/\hbar$ (gravity emerges from the triple).
- $(\ell, \tau, \hbar)$ is the irreducible triple — one process, three aspects.
- Oscillation is a logical inevitability (cannot be removed without destroying Q).
- Gravity is a phase-gradient phenomenon; attraction dominates, repulsion is dark energy.
- Gravity manipulation is not possible (QG-023→031).
- RAR derivation $g_\dagger = cH_0/(2\pi)$ with zero free parameters.
- Parameter compression SM+GR (∼26) → TQM (2+3), ∼5–8× reduction.
- Novel predictions: $w(z) = -1 + 0.015(1+z)^{3/2}$; evolving $g_\dagger(z)$; Euclid sensitivity.

Appendix E

Development Ledger (Selected)

This appendix records, in compact form, a selection of the experiments and audits that underpin the narrative of Chapter 3. It is not exhaustive: the full laboratory book lives in the project's documentation. Each entry states the identifier, the question or claim, and the recorded outcome. Entries are grouped by era.

Identifier	Question / Claim	Outcome / Disposition
Early oscillator experiments		
TQM-001	Does synchronization emerge?	Yes: order parameter $R \to 1$.
TQM-002	Do random matrices give emergence?	No: Wigner spectrum, no structured emergence.
TQM-003	Is static topology sufficient?	No: dynamics outweigh static structure.
TQM-004	Does a field synchronize two asymmetric oscillators?	No: field acts as a resonance medium.
TQM-005	Do resonance clusters form?	Yes: two clusters of three oscillators, $\sim 5\%$ energy participation.
TQM-006	Is there a critical resonance density?	Yes: $\rho_c \approx 0.09$, percolation-like transition.
TQM-007	How many resonance families exist?	Five reported.
TQM-008	Which families are reproducible?	Only F4 (74%, score 0.721); the rest are seed artifacts.
Alternative foundations (ResearchX)		
X001	Hidden-assumption audit	11 found; the static graph is #1.
X002–X004	Do dynamic graphs innovate?	No: node motion, mobility, growth add nothing open-ended.
X005–X006	Does nonlinearity preserve the quantum reading?	No: eigenmodes break into solitons; linear-algebra artifacts.
X007–X008	Universal species theory	16 carrier classes across five regimes.
X009–X015	Deepest foundations	Reversibility + self-consistency is minimally sufficient; Rev$\cap$SC = QM.
X016–X017	Reality map	(R, S) is a map, not a dynamical flow.
X018–X022	Complexity staircase	Level 6 not observed; operator evolution needed.
X023–X027	Open-ended evolution search	Delayed saturation; pigeonhole proof for finite systems.
X028–X029	Finite-universe consequences	Saturation physically irrelevant; hybrid architectures optimal.

Identifier	Question / Claim	Outcome / Disposition
X030–X031	Quantum optimality	QM is the unique global maximum of finite complexity.
X032–X034	Unification	L_Q is not fundamental; emergence gap closed.
X035–X036	Origin of Q; complexity theorem	Q irreducible; Schrödinger derived (classification C).
X037–X039	Born, measurement, randomness	Born derived ($\alpha = 2$); Many-Worlds incompatible; randomness irreducible.
X040–X042	Time, gravity, dimensionality	Time = partial order; gravity = causal set; 3+1 derived.
X043–X046	Fundamental scales	$G = \ell^2 c^3/\hbar$; $\hbar = 1$; $c, G, \hbar$ unified; $\Lambda \sim H^2$.
X047–X049	Matter and gauge structure	Particles = defects; gauge structure emerges; selection gap.
X051–X056, X059–X060	Particle physics	Generations, masses, mixing, neutrinos; SM group preferred not derived.
X057–X058, X060b–X060f	Parameter compression	Two primitives + M^2; M^2 irreducible.
Late phases (140–159)		
Phase 140	Causal-set Λ	$\Lambda \sim 1/\sqrt{N}$ (postdiction).
Phase 144	Radial acceleration relation	$g_\dagger = cH_0/2\pi$; the 2π is a numerical accident.
Phase 148	Flavor/Yukawa reducibility	Koide $Q = 2/3$ contingent, not derived.
Phase 149	Gauge origin	$U(1)$ derived; $SU(2)$ emergent; $SU(3)$ contingent.
Phase 150	Multiplicity "3"	Spacetime 3 derived; internal 3 selected.
Phase 151	Upper bound $N \leq 3$	Empirical/contingent; no stability/anomaly/representation bound.
Phase 152	Random Actualization ensembles	Four independent ensembles under a log-normal law.
Phase 153	Cascade unification	Three universality classes are independent cascades.
Phase 155	Neutrino-Koide	Falsified ($Q_{\max} = 0.585/0.500 < 2/3$).
Phase 159	Koide closure	Seven routes, six no-gos, one blocked; closed question.
QG-080–100	Causal-set foundational program	$\Lambda \sim H^2/c^2$; $a_0 \sim cH$; terminal answer reached.
Hostile review (Round 1)	v1.0 paper audit	1 valid, 11 partially valid, 0 invalid; verdict NOT READY.
Hostile review (Round 2)	FATAL issues audit	6 resolved, 6 partially resolved, 0 open.
v1.0 release	Version readiness	Four dispositions; v1.0 released.

Appendix F

Glossary

Term	Meaning
Q	principle of individuation (ontology)
Random Actualization	ontological chance (assumption A-03)
$(\ell, \tau, \hbar)$	spacetime scale, clock, action (units)
M^2	nonlinearity parameter (single continuous)
L_Q	graph Laplacian $D - A$
L_W	weighted Laplacian $D_K - K$
BDG	Benincasa–Dowker–Glaser d'Alembertian
Δ_g	Laplace–Beltrami operator
$G_{\mu\nu}$	Einstein tensor $R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}$
DERIVED	computable from primitives by theorem
REAL-UNDERIVED	real structure, underivable origin
DRAWN	coincidental draw of the abundance law
T-08. . . T-12	conditional no-go theorems
A-03, A-06, A-07, A-10	assumptions (random actualization; Z_2 lift; gauge dynamics; N/n split)

Appendix G

References

Bibliography

[1] D. B. Malament, "The class of continuous timelike curves determines the topology of spacetime," *J. Math. Phys.* **18**, 1399 (1977).

[2] S. W. Hawking, A. R. King, P. J. McCarthy, "A new topology for curved space–time which incorporates the causal, differential, and conformal structures," *J. Math. Phys.* **17**, 174 (1976).

[3] J. Myrheim, "Statistical geometry," CERN-TH-2538 (1978).

[4] D. M. T. Benincasa, F. Dowker, "Scalar curvature of a causal set," *Phys. Rev. Lett.* **104**, 181301 (2010); L. Glaser, *Class. Quant. Grav.* **31**, 095007 (2014).

[5] A. M. Gleason, "Measures on the closed subspaces of a Hilbert space," *J. Math. Mech.* **6**, 885 (1957).

[6] R. D. Sorkin, "Causal sets: discrete gravity," *Proceedings of the Valdivia Summer School* (2003).

[7] M. Belkin, P. Niyogi, "Laplacian eigenmaps for dimensionality reduction and data representation," *Neural Computation* **15**, 1373 (2003).

[8] Y. Koide, "Fermion mass relation and the generation structure," *Phys. Rev. D* **28**, 252 (1983) [letter 1981].

[9] J. Bertrand, "Théorème relatif au mouvement d'un point attiré vers un centre fixe," *C. R. Acad. Sci.* **77**, 849 (1873).